A Hierarchy of Risk Certificates for Forecasting Representations

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Abstract

We study lower bounds on the prediction risk achievable from a fixed representation with a prescribed decoder sensitivity. A pairwise obstruction is inexpensive to compute but can miss almost all of the optimal risk. For scalar targets, we introduce a hierarchy of normalized minimum-risk problems on blocks of observations. Its size-two member equals the pairwise witness; its expectations increase to the optimal Lipschitz risk under boundedness and compactness. Exact formulas characterize rare conflicts and noisy finite supports, showing why no fixed block size is universally sufficient. Independent-block calibration and stratification with estimated masses yield finite-sample lower certificates. Combined with upper risk bounds, these can certify the minimum feasible cost among declared representation classes. Experiments distinguish population improvement from finite-sample uncertainty and compare against full-sample empirical-minimum calibration. A controlled nonlinear forecasting study tests these decisions with learned Kernel Affine Hull Machine dictionaries and independent confirmation data. Broader synthetic, pen-trajectory, and speech studies show task-dependent accuracy and sensitivity to training-fitted input scaling, including scaling to $[-1, 1]$. The guarantees concern expected squared error under an explicit sampling law; informative natural-data class exclusion remains limited.

Keywords: Kernel Affine Hull Machines, time-series forecasting, soft state abstraction, representation refinement, prediction risk certification

1 Introduction

A forecasting representation maps an observed history to features from which a decoder predicts future observations. Choosing the least costly representation that meets an error tolerance requires two kinds of evidence: an upper risk bound for an accurate fitted predictor, and lower risk bounds that exclude cheaper representation classes. The second question concerns every admissible decoder, including those that have not been fitted. We study these decisions for expected squared error on a fixed observed signal, a declared decoder sensitivity budget, and a common evaluation distribution.

A basic obstruction arises when two targets are farther apart than the sensitivity budget permits their predictions to be. Related pairwise Lipschitz regression arguments are established (Akash et al., 2021). Averaging the squared positive part of this discrepancy over conditionally independent pairs yields a risk lower bound. This inexpensive statistic can be exact, but it can also miss almost all of the attainable class risk. The central question of this paper is how to understand and reduce that population slack without confusing it with uncertainty from a finite calibration sample.

Our main construction replaces pairs by blocks of observations. Within each block, we minimize squared error over predictions satisfying all pairwise Lipschitz constraints and normalize by one less

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than the block size. The size-two statistic is exactly the pairwise obstruction. For scalar targets, the expectations of these statistics form a nondecreasing hierarchy of lower bounds. Under unconditional iid sampling, bounded targets, and compact representation support, they converge to the optimal Lipschitz risk. Larger blocks capture compatibility constraints that individual pairs cannot express. They also require more optimization and leave fewer independent block scores for calibration. Population improvement therefore does not imply a uniformly stronger finite-sample certificate.

Conditional pairing provides a complementary way to expose an obstruction. Comparisons within a predefined history stratum can concentrate on distinctions that global pairing rarely encounters. Reweighting by stratum probabilities preserves the original marginal risk. We characterize the pair witness's slack and give sharp examples and counterexamples, then account for both random stratum counts and estimated masses in calibration. The concentration inequalities and simultaneous-selection principles used for calibration are established; the contribution lies in the hierarchy, its relationship to pairing, and the representation decisions that the resulting lower information can support.

Kernel Affine Hull Machines (KAHMs) provide a concrete forecasting application. Their established local reconstruction models yield soft associations between a history and reference groups (Kumar et al., 2024, 2025; Kumar and Kargaran, 2026). We retain these models and append feature blocks under a sum-of-block-norms decoder budget. This preserves earlier predictors and makes exact block duplication immaterial to the feasible class. The analysis requires neither an exact linear feature evolution nor a Markov representation. Its risk concerns observed future outputs. Preprocessing, including training-fitted scaling to $[-1, 1]$, forms part of the representation and is frozen before calibration.

The paper makes three contributions.

1. **A hierarchy and its population limits.** We connect the pair witness to normalized minimum-risk problems on larger blocks, establish monotonicity and consistency under the stated conditions, and identify exact and arbitrarily loose pairwise cases.

2. **Calibration and representation decisions.** Independent block scores and valid stratification give class-wide lower certificates. Combined with fitted-risk upper bounds and retained-block geometry, they support accuracy, improvement, and minimum feasible cost statements for a frozen family of alternatives.

3. **Tests of usefulness and failure.** Exact-risk comparisons separate population slack, statistical precision, and optimization cost. A controlled study tests class exclusion and refinement using learned KAHM dictionaries and fresh independent calibration episodes. Broader forecasting studies on six synthetic systems, pen trajectories, and speech energy assess accuracy, scaling, and the limits of calibration.

The evidence supports a qualified use of the certificates. Empirical-minimum calibration is often tighter than pairwise calibration. Useful class exclusion on a constructed mechanism does not establish its prevalence in natural forecasting tasks. KAHM accuracy and the effect of input scaling depend on the task and decoder; conventional regressors remain stronger in several studies. All confidence statements concern expected risk under an explicit sampling law, with independent calibration units and bounded target losses.

2 Related work

Geometric representations and Koopman prediction. KAHMs learn reconstruction models in reproducing kernel Hilbert spaces (RKHSs) (Kumar et al., 2024, 2025). We use these established reconstruction models to construct forecasting features. Soft state aggregation (Singh et al., 1994), approximate state abstraction (Abel et al., 2016), and task-oriented latent transition models (Gelada et al., 2019) likewise distinguish a representation from its predictive use. Extended dynamic mode decomposition approximates Koopman operators on finite dictionaries (Williams et al., 2015); learned dictionaries (Li et al., 2017) and RKHS regression bounds (Kostic et al., 2022) provide related forecasting frameworks. Basis-invariant consistency diagnostics (Haseli and Cortés, 2023), multistep projection-error guarantees and refinement (Conradie et al., 2026), and residual-based dictionary learning (Coote and Colbrook, 2026) address dictionary quality. Our decisions instead compare expected observed-output error with a declared tolerance while preserving a common target and sampling law.

Closest KAHM prediction-limit analysis. Kumar and Kargaran (2026) study norm-bounded linear prediction of successor soft coordinates, centered lower certificates, empirical-risk-based alternatives, fitted-risk upper bounds, and independent-trajectory evaluation. We use their KAHM reconstruction and sampling principle. Our analysis concerns observed-output error and a positive-part pair witness. It characterizes the witness's slack, extends it to a monotone block hierarchy converging to the scalar Lipschitz optimum under compactness, studies stratification, and derives cost certificates for classes that preserve earlier feature blocks. The class is unchanged when an existing block is duplicated. Independent-episode calibration and acceptance based on an upper risk bound are established components of this framework. Our experiments compare both centered lower bounds and lower bounds obtained by certified empirical-risk minimization.

Pairwise bounds and Lipschitz regression. Akash et al. (2021) derive a Lipschitz-constrained mean squared error (MSE) bound using paired residuals and target separation. Their argument is a close antecedent to the geometric step in our witness; our contribution concerns its conditional positive-part form, sharpness and slack, and statistically valid stratification under the unchanged marginal risk. Finite-support Lipschitz regression can be posed as a constrained optimization problem (Gottlieb et al., 2017); we use this class directly for an empirical-minimum comparator. Model-class falsification (Müller et al., 2025) and minimal-empirical-risk bounds (Wu and Yang, 2026) provide broader context. The Gamma test uses local pair differences to estimate regression noise (Evans and Jones, 2002). Lipschitz set-membership identification (Milanese and Novara, 2004), trajectory invalidation (Jin et al., 2020), and corruption-robust prediction ranges (Blum and Nguyen, 2026) concern related feasibility questions with different sampling and error criteria.

Empirical minima and optimization over blocks. Expected sample-average optimization values and repeated-batch lower bounds are established in stochastic programming (Linderoth et al., 2006; Guigues et al., 2017). Optimization of empirical U-statistics and the tradeoff between computation and statistical precision are also established (Clémençon et al., 2016). Our block objective can be written as the minimum of a residual-difference U-statistic. We do not claim that empirical-minimum lower bounds or block averaging are new. The specific result is the centered normalization for scalar Lipschitz regression: its size-two member is the pair witness, its expectations increase to the class optimum under compactness, and exact examples quantify why a fixed block size can remain inadequate. Calibration of the resulting scores uses standard bounded-mean inequalities.

Calibration, refinement, and dependence. We use classical bounded-mean inequalities (Hoeffding, 1963) and empirical Bernstein and standard-deviation bounds (Maurer and Pontil, 2009). The simultaneous testing principle follows Learn then Test (Angelopoulos et al., 2025). Our group norm is established in group selection and multiple-kernel learning (Bach, 2008); here its max-block dual metric gives a comparable sensitivity budget as dictionaries grow. Soft gating and residual fitting also have established precedents (Jordan and Jacobs, 1994; Bühlmann and Yu, 2003). The additional cost conclusion comes from jointly excluding cheaper classes and accepting a fitted model.

Within-episode dependence is allowed because concentration is applied to independent episode summaries. A chronological split of one dependent record does not provide that independence. Forecasting bounds under other sampling assumptions (McDonald et al., 2017) and dependence-aware conformal analysis (Oliveira et al., 2024) illustrate the additional conditions needed; their guarantees are not consequences of the present theorem.

3 Pairwise obstructions and their limitations

This section establishes lower bounds on the risk achievable from a fixed representation, upper bounds for fitted forecasts, and conditions for representation refinement. All results refer to the same distribution of histories and observed forecasting targets. In particular, they do not require that the histories form a Markov state, that the trajectory distribution is stationary, or that the learned coordinates satisfy an exact linear evolution equation.

3.1 A directly estimable pairwise obstruction

Let $X \in \mathcal{X}$ be a random available history, let $Y \in \mathbb{R}^q$ be its future target, and let $U = \Gamma(X) \in \mathcal{U}$ be a fixed, measurable representation. Here $\mathcal{X}$ and $\mathcal{U}$ denote the history and representation spaces. A measurable pseudometric $d : \mathcal{U} \times \mathcal{U} \to [0, \infty)$ is symmetric, vanishes on the diagonal, and satisfies the triangle inequality; unlike a metric, it may assign zero distance to distinct points. For a sensitivity budget $\kappa \geq 0$, define

$$\mathcal{G}_\kappa(d) = \{g : \mathcal{U} \to \mathbb{R}^q \text{ measurable} : \|g(u) - g(v)\|_2 \leq \kappa d(u, v) \text{ for all } u, v \in \mathcal{U}\},$$
$$R^*_\kappa(\Gamma, d) = \inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}\|Y - g(\Gamma(X))\|_2^2.$$

Thus κ limits how much a forecast can change when its representation changes, and R^*_κ is the best population squared-error risk permitted by that limit. For any predictor $f : \mathcal{X} \to \mathbb{R}^q$, write $R(f) = \mathbb{E}\|Y - f(X)\|_2^2$. The representation, pseudometric, and budget are fixed before the calibration data are observed. The class $\mathcal{G}_\kappa(d)$ may contain more decoders than the fitted family; a lower bound on its optimal risk therefore also applies to that family. When the representation and pseudometric are clear, we abbreviate $R^*_\kappa(\Gamma, d)$ to R^*_κ and similarly suppress the arguments of the witness defined below.

Any training-fitted preprocessing is part of Γ. The population results below do not require the input coordinates to lie in $[-1, 1]$. For the finite-sample certificates, the deterministic range assumption concerns the target and prediction loss. A common value of κ across preprocessing choices limits sensitivity in each resulting feature metric; it does not impose a common Lipschitz constant in the original input units. Each choice therefore specifies its own representation class and must be frozen before calibration.

To define the pairwise statistic, first draw a conditioning variable A. Then draw (X, Y) and (X', Y') independently from the same conditional law given A, and set $U = \Gamma(X)$ and $U' = \Gamma(X')$. Each copy must have the original marginal law of (X, Y). Taking A to be constant gives an ordinary independent pair. With $z_+ = \max\{z, 0\}$, define

$$W_{\kappa,A}(\Gamma, d) = \frac{1}{2}\mathbb{E}\left[\left(\|Y - Y'\|_2 - \kappa d(U, U')\right)_+^2\right]. \tag{1}$$

The expectation includes both the draw of A and the two conditional draws. We call $W_{\kappa,A}$ a *pair witness*, or *obstruction*: it measures target separation that a decoder with budget κ cannot reproduce from the representation separation. The variable A specifies a sampling construction. Arbitrary nearest-neighbor matching need not have this conditional independence property.

Theorem 1 (Pairwise prediction obstruction) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$. Then*

$$R_\kappa^*(\Gamma, d) \geq W_{\kappa,A}(\Gamma, d). \tag{2}$$

For a representation space $\mathcal{U} \subseteq \mathbb{R}^p$ with $d(u, v) = \|u - v\|_2$, where p is its ambient dimension, assume also $\mathbb{E}\|U\|_2^2 < \infty$. Define

$$\sigma_{Y|A}^2 = \mathbb{E}\|Y - \mathbb{E}[Y \mid A]\|_2^2, \qquad \tau_{U|A}^2 = \mathbb{E}\|U - \mathbb{E}[U \mid A]\|_2^2,$$

where $\sigma_{Y|A}$ and $\tau_{U|A}$ are the nonnegative square roots. These are scalar, averaged conditional variances, rather than functions of a particular value of A. Then

$$W_{\kappa,A}(\Gamma, \|\cdot\|_2) \geq (\sigma_{Y|A} - \kappa\tau_{U|A})_+^2. \tag{3}$$

Both inequalities use the same conditioning variable and evaluation law.

Proof Fix $g \in \mathcal{G}_\kappa(d)$ with finite risk and define the two residuals $r = Y - g(U)$ and $r' = Y' - g(U')$. Since $Y - Y' = r - r' + g(U) - g(U')$, the triangle inequality gives

$$\|Y - Y'\|_2 \leq \|r - r'\|_2 + \kappa d(U, U'), \qquad (\|Y - Y'\|_2 - \kappa d(U, U'))_+ \leq \|r - r'\|_2.$$

Conditional on A, the residuals are independent and identically distributed. Expanding the squared norm therefore yields

$$\begin{aligned}\mathbb{E}[\|r - r'\|_2^2 \mid A] &= 2\mathbb{E}[\|r\|_2^2 \mid A] - 2\mathbb{E}[r \mid A]^\top \mathbb{E}[r' \mid A] \\ &= 2\left\{\mathbb{E}[\|r\|_2^2 \mid A] - \|\mathbb{E}[r \mid A]\|_2^2\right\}.\end{aligned}$$

Square the preceding pointwise bound, take expectations, and divide by two. This gives $W_{\kappa,A} \leq \mathbb{E}\|r\|_2^2 = R(g \circ \Gamma)$. Taking the infimum over g proves (2); predictors with infinite risk satisfy the bound trivially.

For (3), set $a = \|Y - Y'\|_2$ and $b = \|U - U'\|_2$. Pointwise, $a \leq (a - \kappa b)_+ + \kappa b$. For a scalar random variable V, the notation $\|V\|_{L^2} = (\mathbb{E}V^2)^{1/2}$ denotes its root second moment under the pairing law. The triangle inequality in L^2 gives

$$\|(a - \kappa b)_+\|_{L^2} \geq (\|a\|_{L^2} - \kappa\|b\|_{L^2})_+.$$

The same conditional-independence calculation gives $\mathbb{E}a^2 = 2\sigma_{Y|A}^2$ and $\mathbb{E}b^2 = 2\tau_{U|A}^2$. Substitute these identities, square, and divide by two to obtain (3). ■

Proposition 2 (An exact decomposition of witness slack) *For any admissible decoder g with finite risk, put $r = Y - g(U)$, $r' = Y' - g(U')$, and $v = (\|Y - Y'\|_2 - \kappa d(U, U'))_+$. Then*

$$R(g \circ \Gamma) - W_{\kappa,A} = \mathbb{E}\|\mathbb{E}[r \mid A]\|_2^2 + \frac{1}{2}\mathbb{E}\big[\|r - r'\|_2^2 - v^2\big]. \tag{4}$$

Both terms are nonnegative. Consequently, g attains the witness, meaning $R(g \circ \Gamma) = W_{\kappa,A}$, if and only if $\mathbb{E}[r \mid A] = 0$ almost surely and $\|r - r'\|_2 = v$ almost surely. Whenever $r \neq r'$, the latter condition requires $\|g(U) - g(U')\|_2 = \kappa d(U, U')$. It also requires the decoder difference to be a nonnegative scalar multiple of the residual difference; the scalar may be zero.

Proof Conditional independence gives $\frac{1}{2}\mathbb{E}\|r - r'\|_2^2 = R(g \circ \Gamma) - \mathbb{E}\|\mathbb{E}[r \mid A]\|_2^2$. Subtract the witness and use $v \leq \|r - r'\|_2$ from the preceding proof. Equality requires both nonnegative terms to vanish. If $r \neq r'$, it also requires equality in $\|Y - Y'\|_2 \leq \|r - r'\|_2 + \|g(U) - g(U')\|_2 \leq \|r - r'\|_2 + \kappa d(U, U')$, giving the stated conditions. ■

The witness directly estimates a squared distance discrepancy. It does not estimate two variances separately and then subtract their square roots. This distinction can matter when little input variation remains in the representation. The certificates below use concentration for bounded means; they establish no new minimax estimation rate. Taking a square root of a risk bound can still produce slow convergence near zero. Moreover, (3) compares certificates under a common pairing law. It does not compare pairing at a common time origin with conditioning on a different grouping, such as a reference-class label $c(X)$, where c assigns a history to a group.

Proposition 3 (Exactness on a two-atom law) *Suppose (U, Y) takes only two values: (u_0, y_0) with probability $p \in (0, 1)$ and (u_1, y_1) with probability $1 - p$. Use Euclidean representation distance and an unconditioned independent pair, and omit the constant conditioning variable from the notation. Then*

$$R_\kappa^* = W_\kappa = p(1 - p)\big(\|y_1 - y_0\|_2 - \kappa\|u_1 - u_0\|_2\big)_+^2. \tag{5}$$

An affine map attains this optimum. If forecasts must lie in a nonempty closed convex set containing y_0, y_1, projecting that map onto the set preserves both the optimum and the Lipschitz budget.

Proof Write $a = \|y_1 - y_0\|_2$, $b = \|u_1 - u_0\|_2$, and $\mu = py_0 + (1 - p)y_1$. The unequal ordered pairs have total probability $2p(1 - p)$, whereas equal pairs contribute zero. Thus (1) gives the right side of (5) as a lower bound.

To attain the bound, first consider $a = 0$: the constant forecast y_0 has zero risk. If $a > 0$, let $v = (y_1 - y_0)/a$ be the unit target direction and let $s = \min\{a, \kappa b\}$ be the allowed fitted separation. Choose

$$f_0 = \mu - (1 - p)sv, \qquad f_1 = \mu + psv.$$

Then $y_0 - f_0 = -(1 - p)(a - s)v$ and $y_1 - f_1 = p(a - s)v$, so their weighted squared error is $p(1 - p)(a - s)^2$, exactly the proposed bound. If $b = 0$, then $s = 0$ and the constant map μ realizes both fitted values. If $b > 0$, use

$$g(u) = f_0 + \frac{\langle u - u_0, u_1 - u_0\rangle}{b^2}(f_1 - f_0).$$

Its linear part has operator norm $\|f_1 - f_0\|_2/b = s/b \le \kappa$, so it is an admissible affine decoder. Both fitted values lie on the segment joining y_0 and y_1. Projection onto the stated convex set fixes them and is nonexpansive, meaning that it cannot increase distances. ■

Exactness also holds on continuous supports. Let T be a square-integrable real random variable, set $U = T$ and $Y = b + \beta T v$, and assume $b, v \in \mathbb{R}^q$, $\|v\|_2 = 1$, and $\beta \ge 0$. Here β is the target's slope, and $\mathrm{Var}(T) = \mathbb{E}(T - \mathbb{E}T)^2$. For independent T, T' and $d(t, t') = |t - t'|$, the pair discrepancy is $(\beta - \kappa)_+|T - T'|$. Since $\mathbb{E}(T - T')^2 = 2\,\mathrm{Var}(T)$, its witness is $(\beta - \kappa)_+^2\,\mathrm{Var}(T)$. The affine decoder

$$g(t) = b + \beta\mathbb{E}[T]v + \min\{\beta, \kappa\}(t - \mathbb{E}[T])v$$

has Lipschitz constant at most κ and residual $(\beta - \kappa)_+(T - \mathbb{E}T)v$. Its risk equals the witness, so $W_\kappa = R^*_\kappa$ also in this continuous-support example.

Two sources of population looseness. Even with independent unconditioned pairs, the witness need not approximate the optimum by a fixed relative factor. Let $\kappa = 1$ and let (U, Y) take values $(3, 0), (0, -1), (1, 1)$ with probabilities $1 - 2\varepsilon, \varepsilon, \varepsilon$, where $0 < \varepsilon < 1/2$. Only the two rare atoms have a positive discrepancy, so $W_\kappa = \varepsilon^2$. The fitted values $g(0) = -1/2$, $g(1) = 1/2$, and $g(3) = 0$ attain risk $\varepsilon/2$. Linear interpolation between these points, with constant values outside $[0, 3]$, defines a 1-Lipschitz decoder. Conversely, any admissible fitted values $v_0 = g(0)$ and $v_1 = g(1)$ satisfy $|v_1 - v_0| \le 1$. The minimum of $(-1 - v_0)^2 + (1 - v_1)^2$ under this constraint is $1/2$, attained at $v_0 = -1/2$ and $v_1 = 1/2$. The two rare atoms alone therefore contribute at least $\varepsilon/2$ to every decoder's risk. Thus $R^*_\kappa = \varepsilon/2$ and $W_\kappa/R^*_\kappa = 2\varepsilon$ can approach zero. The same example lies in Δ_2 after mapping u to $(u/3, 1 - u/3)$ and setting $\kappa = 3/\sqrt{2}$. Stratification below replaces the quadratic probability of encountering this rare pair by the probability of its stratum.

A separate gap comes from comparing the fitted affine family with all Lipschitz decoders. For equally likely $X \in \{-1, 0, 1\}$, take $\Phi(x) = ((x + 1)/2, (1 - x)/2)^\top$, $Y = X^2$, and $\kappa = \sqrt{2}$, with Euclidean distance between feature vectors. The target satisfies the Lipschitz inequalities on this support and has a scalar Lipschitz extension, so $W_\kappa = R^*_\kappa = 0$. An affine function of $\Phi(x)$ is affine in x; clipping it to $[0, 1]$ preserves monotonicity. A nondecreasing forecast cannot fit the target pair $(1, 0)$ at $x = -1, 0$: its minimum sum of squared errors on that pair is $1/2$, attained when both fitted values are $1/2$. A nonincreasing forecast has the same obstruction on the target pair $(0, 1)$ at $x = 0, 1$. Averaging over the three atoms gives MSE at least $1/6$ for every clipped affine forecast. The unprojected affine optimum is the constant $2/3$, with MSE $2/9$. Thus a sharp lower bound for all Lipschitz decoders need not describe the optimum of a restricted affine family.

4 A hierarchy of consistent block witnesses

The pair witness can lose information because its optimal fitted values are chosen separately for each pair. Requiring one set of fitted values to respect every distance in a larger block removes some of this freedom. We now make this observation precise for a *scalar* target. The result connects the pair obstruction to the optimal Lipschitz risk, while exposing the statistical cost of using larger blocks.

Fix the representation, pseudometric, and sensitivity budget from Section 3. Let $(U_i, Y_i)_{i=1}^b$ be independent copies from the full evaluation law, where $b \ge 2$, and assume $Y \in [y_-, y_+]$ almost

surely. Write $D = y_+ - y_-$. Define

$$T_b = \frac{1}{b-1} \min_{v \in \mathbb{R}^b} \sum_{i=1}^{b} (Y_i - v_i)^2, \qquad \mathcal{W}_b = \mathbb{E}T_b, \\ \text{subject to } |v_i - v_j| \le \kappa d(U_i, U_j) \quad (i, j \le b). \tag{6}$$

The feasible set contains every constant vector. The minimum therefore exists and is unique. Its value is unchanged if the fitted values are restricted to $[y_-, y_+]$, because clipping preserves feasibility and cannot increase squared error. The factor $1/(b-1)$ is essential: if a_b denotes the usual minimum empirical MSE, then $T_b = ba_b/(b-1)$. This correction is justified by the freedom to translate every fitted value by the same constant; it is not a generic correction for arbitrary empirical-risk minima.

Theorem 4 (Monotone block hierarchy) *Under these assumptions, the following statements hold.*

(i) The first member is exactly the pair score:

$$T_2 = \tfrac{1}{2}\big(|Y_1 - Y_2| - \kappa d(U_1, U_2)\big)_+^2.$$

Consequently $\mathcal{W}_2 = W_{\kappa,A}$ when A is constant.

(ii) For every $b \ge 2$,

$$0 \le T_b \le B_b^{\mathrm{blk}}, \qquad B_b^{\mathrm{blk}} = \frac{D^2 \lfloor b^2/4 \rfloor}{b(b-1)}, \qquad \mathcal{W}_b \le R_\kappa^*. \tag{7}$$

(iii) If $2 \le m \le b$ and $T_m(S)$ is the score computed from a subset $S \subseteq \{1, \dots, b\}$ of size m, then

$$\binom{b}{m}^{-1} \sum_{|S|=m} T_m(S) \le T_b. \tag{8}$$

In particular, $\mathcal{W}_2 \le \mathcal{W}_3 \le \cdots \le R_\kappa^*$.

(iv) If the representation takes values in a compact metric space and d is that metric, then $\mathcal{W}_b \uparrow R_\kappa^*$ as $b \to \infty$.

Proof For two observations, let $\Delta = |Y_1 - Y_2|$ and $c = \kappa d(U_1, U_2)$. If $\Delta \le c$, both targets can be fitted exactly. Otherwise the optimal fitted separation is c, with the two residuals having magnitude $(\Delta - c)/2$ and opposite signs. Their squared sum is $(\Delta - c)^2/2$, proving (i).

For any fixed admissible decoder g of finite risk, put $r_i = Y_i - g(U_i)$ and $\bar{r} = b^{-1} \sum_i r_i$. The translated values $v_i = g(U_i) + \bar{r}$ are feasible. Thus

$$T_b \le \frac{1}{b-1} \sum_i (r_i - \bar{r})^2, \qquad \mathcal{W}_b \le \mathrm{Var}(Y - g(U)) \le R(g \circ \Gamma).$$

Taking the infimum over g proves the population bound. Using the constant fitted value $\bar{Y}$ instead bounds T_b by the unbiased sample variance of the targets. This variance is a convex function of each target coordinate, so its maximum over $[y_-, y_+]^b$ occurs at interval endpoints. If k coordinates equal

y_+ and $b-k$ equal y_-, its value is $D^2k(b-k)/\{b(b-1)\}$. Maximizing over integer k proves the range bound.

For (iii), let v minimize the full-block objective and set $r_i = Y_i - v_i$. Translation invariance implies $\sum_i r_i = 0$: otherwise translating v by the residual mean would strictly lower the objective. Hence

$$T_b = \frac{1}{b(b-1)} \sum_{i<j} (r_i - r_j)^2. \tag{9}$$

On a subset S, restricting v and translating by that subset's residual mean gives feasible fitted values. Therefore

$$T_m(S) \le \frac{1}{m(m-1)} \sum_{i<j,\ i,j\in S} (r_i - r_j)^2.$$

A uniformly chosen m-subset contains a given pair with probability $m(m-1)/\{b(b-1)\}$. Averaging proves (8); taking expectations proves monotonicity. This is an average-over-subsets statement, not an ordering between arbitrary realized blocks of different sizes.

To prove (iv), any feasible vector extends to a scalar Lipschitz function by $g_v(u) = \min_i\{v_i + \kappa d(u, U_i)\}$. The triangle inequality and the finite constraints show that this function is κ-Lipschitz and equals v_i at U_i. Clipping it to $[y_-, y_+]$ preserves those properties. Thus a_b is the empirical minimum over the same bounded Lipschitz class as the population problem. On a compact metric space this class is compact in the uniform norm: it is uniformly bounded and equicontinuous. For each fixed $\epsilon > 0$, a finite uniform ϵ-net and the ordinary strong law give uniform convergence of empirical risks over that net. Since $|(y - f(u))^2 - (y - g(u))^2| \le 2D\|f - g\|_\infty$ for interval-valued f, g, the convergence extends to the entire class. Therefore $a_b \to R^*_\kappa$ almost surely. Multiplication by $b/(b-1)$ and the bound $T_b \le D^2/2$ give $\mathcal{W}_b \to R^*_\kappa$ by dominated convergence. ■

The result strengthens the population witness without changing the decoder class. In particular, every block score dominates the average of its pair scores. The additional information is joint consistency of the fitted values across pairs, rather than a new concentration inequality. Optimization of empirical U-statistics is established (Clémençon et al., 2016): equation (9) places this construction in that framework. Lower bounds obtained by averaging empirical optimization values over independent batches are also established in stochastic programming (Linderoth et al., 2006; Guigues et al., 2017). The specific connection here is the centered normalization, its exact pair-obstruction member, and the monotone hierarchy for the Lipschitz prediction problem.

4.1 Exact gaps and the limits of fixed block sizes

Consistency for a fixed distribution does not provide a universal choice of b. Two exact calculations distinguish rare predictive conflicts from the capacity to interpolate many distinct inputs.

Proposition 5 (Two exact block-size laws) *For the three-atom distribution from Section 3, with* $(U, Y) = (3, 0), (0, -1), (1, 1)$ *of probabilities* $1 - 2\varepsilon, \varepsilon, \varepsilon$ *and* $\kappa = 1$*,*

$$R^*_\kappa = \frac{\varepsilon}{2}, \qquad \mathcal{W}_b = \frac{2b\varepsilon - 1 + (1 - 2\varepsilon)^b}{4(b-1)}. \tag{10}$$

For every fixed b*,* $\mathcal{W}_b / R^*_\kappa \to 0$ *as* $\varepsilon \downarrow 0$*.*

Separately, let U be uniform on $M \geq 1$ points whose pairwise distances are at least D/κ, where $D, \kappa > 0$, and let $Y \in \{0, D\}$ be independent of U and equiprobable. Then

$$R^*_\kappa = \frac{D^2}{4}, \qquad \mathcal{W}_b = \frac{D^2}{4(b-1)}\left[b - M\{1-(1-1/M)^b\}\right]. \tag{11}$$

For fixed b, $\mathcal{W}_b \to 0$ as $M \to \infty$, while the optimal risk stays $D^2/4$.

Proof In the first law, let N_- and N_+ count the two rare atoms in a block, and put $N = N_- + N_+$. The common atom can be fitted at zero without restricting the optimum at the rare atoms. Their constrained weighted least-squares error is N_-N_+/N when $N > 0$, and zero otherwise. Conditional on N, N_- is binomial with parameters N and $1/2$, so the conditional mean of this error is $(N-1)/4$ for $N \geq 1$. Since $N \sim \mathrm{Binomial}(b, 2\varepsilon)$,

$$(b-1)\mathcal{W}_b = \tfrac{1}{4}\{\mathbb{E}N - \Pr(N \geq 1)\} = \tfrac{1}{4}\{2b\varepsilon - 1 + (1-2\varepsilon)^b\}.$$

Its expansion at zero gives $\mathcal{W}_b/R^*_\kappa = b\varepsilon + O(\varepsilon^2)$.

In the second law, any fitted values in $[0, D]$ at distinct support points satisfy the Lipschitz constraints. At a support point observed $n > 0$ times, the empirical optimum is its target mean, with expected squared residual sum $(n-1)D^2/4$. If O_b is the number of occupied support points, the conditional expected total is $(b - O_b)D^2/4$. Each of the M points is occupied with probability $1-(1-1/M)^b$. Taking expectations gives (11). Independence of Y and U gives the population optimum $D^2/4$, attained by the constant $D/2$. ■

The second example consists entirely of finite, hence compact, metric spaces. It rules out a distribution-free additive approximation rate depending only on b and the target range over all such spaces. The support points can be $u_j = De_j/(\kappa\sqrt{2})$ in $\mathbb{R}^M$, where e_j are the coordinate unit vectors. Their diameter is D/κ for every $M \geq 2$, so a bound on representation diameter alone does not remove this limitation. For fixed b, $\mathcal{W}_b/R^*_\kappa = b/(2M) + O(M^{-2})$ as $M \to \infty$. The first example shows why increasing block size can be necessary even on three fixed support points. Larger blocks address these losses, but do not make an uninformative certificate evidence of adequate representation.

4.2 Calibration and certified optimization values

Use disjoint blocks of b independent calibration units, giving $K_b = \lfloor n/b \rfloor$ independent scores. This makes the effect of block size explicit: increasing b tightens $\mathcal{W}_b$ but decreases the number of independently calibrated scores. The hierarchy therefore does not order finite-sample confidence endpoints.

The quadratic program in (6) must be solved from below for a lower certificate. Write its inequalities as $Av \leq c$, where rows of A are coordinate differences and $c_{ij} = \kappa d(U_i, U_j)$. For every nonnegative multiplier vector λ, weak duality gives the lower value

$$d_b(\lambda) = \lambda^\top(AY - c) - \tfrac{1}{4}\|A^\top\lambda\|_2^2 \;\leq\; \min_{Av \leq c} \|Y - v\|_2^2. \tag{12}$$

This follows by minimizing $\|Y - v\|_2^2 + \lambda^\top(Av - c)$ over v; its minimizer is $Y - A^\top\lambda/2$. Thus $\underline{T}_b = \max\{0, d_b(\lambda)/(b-1)\} \leq T_b$. A primal value alone has the wrong inequality direction. The

implementation reports primal feasibility and the primal–dual gap; floating-point tolerances are numerical checks rather than proofs by verified arithmetic.

Corollary 6 (Independent-block lower calibration) *Suppose the same fixed measurable procedure computes each score $\underline{T}_{b,k} \in [0, B_b^{\mathrm{blk}}]$ from block k, with $\underline{T}_{b,k} \le T_{b,k}$, and $K_b \ge 1$. For an allocated failure probability $\alpha \in (0,1)$ and $B_b^{\mathrm{blk}} > 0$, let*

$$L_b = B_b^{\mathrm{blk}} \, \mathrm{kl}_- \left(\frac{K_b^{-1} \sum_{k=1}^{K_b} \underline{T}_{b,k}}{B_b^{\mathrm{blk}}}, \frac{\log(1/\alpha)}{K_b} \right), \tag{13}$$

where kl_- is defined in Appendix B.3. Then $L_b \le \mathcal{W}_b \le R_\kappa^*$ with probability at least $1-\alpha$. If $B_b^{\mathrm{blk}} = 0$, take $L_b = 0$. A finite family of block sizes, representations, horizons, and comparison methods can be selected after calibration when the sum of their allocated failure probabilities is at most the declared total.

Proof The bounded-mean inequality (37) gives $L_b \le \mathbb{E}\underline{T}_{b,1}$ with failure probability at most α. The pointwise score inequality and Theorem 4 give $\mathbb{E}\underline{T}_{b,1} \le \mathcal{W}_b \le R_\kappa^*$. A union bound proves the finite-family assertion; different methods may reuse the same calibration observations. ■

The empirical Bernstein radius can be substituted when $K_b \ge 2$. Block sizes and optimization rules are fixed before calibration; taking the largest endpoint across a frozen family requires the stated joint allocation. This is distinct from multiplying the full-sample empirical-minimum endpoint in Appendix B.4 by $n/(n-1)$, which that endpoint's proof does not justify.

Conditioning and forecasting origins. For a common origin or stratum A, draw the b observations independently conditional on A. The same proof gives a monotone hierarchy $\mathcal{W}_{b,A} \le R_\kappa^*$. For finitely many values of A and the compact-support conditions above, its limit is

$$\sum_a \Pr(A = a) \inf_{g \in \mathcal{G}_\kappa(d)} \mathbb{E}[(Y - g(U))^2 \mid A = a],$$

which can be strictly below the optimum of one shared decoder. The decoder attaining each conditional infimum may differ with a. Consequently, averaging a block score over common forecast origins preserves validity and monotonicity but does not generally establish convergence to the shared-decoder risk. For that consistency claim, one may instead draw an independent uniform origin in each independent episode; the resulting observations are independent draws from the full evaluation law. Neither construction treats overlapping windows as independent calibration units.

5 Learning and certifying KAHM forecasts

The procedure has three stages. KAHM reconstruction models map each history to geometric features; a constrained decoder maps those features to future observations; and independent calibration assesses the resulting forecasts. Keeping these stages separate also allows alternative representations to use the same decoder and evaluation procedure.

5.1 Notation and basic conventions

Vectors are columns unless used as data-matrix rows. We write $\|\cdot\|_2$, $\|\cdot\|_{\mathrm{F}}$, and $\|\cdot\|_{\mathrm{op}}$ for Euclidean, Frobenius, and operator norms. Matrix inner products are $\langle M, N\rangle_{\mathrm{F}} = \mathrm{tr}(M^\top N)$. Logarithms are natural; $z_+ = \max\{z, 0\}$; $\|V\|_{L^2} = (\mathbb{E}\|V\|_2^2)^{1/2}$. The projection $\Pi_{\mathcal{Y}}$ onto a nonempty closed convex output set is nonexpansive; for a box it clips coordinates. All random variables and prediction maps are measurable.

Table 1 collects the main quantities. The forecasting construction and the statistical statements below give their precise domains and sampling assumptions.

5.2 A fixed raw-output forecasting task

Let $y_t \in \mathbb{R}^{d_y}$ be the observation at integer time t, where $d_y \geq 1$ is the number of observed coordinates. For a history length $m \geq 1$, define the column vector

$$x_t = (y_{t-m+1}^\top, \ldots, y_t^\top)^\top \in \mathbb{R}^{md_y}.$$

Covariates available at time t may be appended to this vector. A forecast origin is a time t at which a prediction is made using only this available information. Let $\mathcal{H}$ be a finite, nonempty set of positive integer horizons, ordered increasingly, and let $H = |\mathcal{H}|$ be its size. The joint target is

$$Y_t = (y_{t+h}^\top : h \in \mathcal{H})^\top \in \mathbb{R}^q, \qquad q = d_y H.$$

Thus each training row contains all requested future observations in the same order. The theory also considers one horizon at a time, in which case its target dimension is d_y (or one for a separately scored coordinate).

The history length, any fixed target transformation, horizons, and origin sampling rule define the prediction task. Here "raw output" means the observed signal under that fixed transformation, rather than learned feature coordinates. When comparing history lengths, we use the same origins, starting after the longest history is available. Every candidate is therefore evaluated under the same target and origin law.

The main learner directly predicts each requested horizon, sharing a geometric representation and a joint coefficient budget. It does not generate an entire trajectory by iterating one-step forecasts. The rollout baselines instead iterate a fitted feature-transition model and decode the resulting features. Both approaches are evaluated on the same observed targets; a small one-step feature error alone is not assumed to guarantee an accurate long rollout.

An *episode* is one complete trajectory or other prespecified independent sampling unit. Development data comprise training data, used to fit features and coefficients, and validation data, used to choose hyperparameters or a representative within each model family. Calibration data provide the confidence bounds and may select among candidates frozen before calibration. A separate test set describes forecasting performance after these decisions. Its empirical error estimates population risk; it is not an exact risk value.

5.3 KAHM dictionaries and stable soft associations

A dictionary block describes a history through its associations with a set of reference groups. KAHM associations depend on local reconstruction models. Radial basis function (RBF) associations instead depend on distances to group centers; affine-hull associations use residuals from fitted local affine

Symbol	Meaning
y_t, d_y	Observation at time t and its dimension.
x_t, m	Causal history and number of observed time steps in it.
$\mathcal{H}$, H, q	Set of forecast horizons, its size, and target dimension; for a joint forecast, $q = d_y H$.
Φ_b, C_b	Feature block b and its number of coordinates.
Γ_s, p_s	Stack of the first s feature blocks and total dimension $p_s = \sum_{b=1}^{s} C_b$.
a, W_b	Decoder intercept in $\mathbb{R}^q$ and block coefficients in $\mathbb{R}^{C_b \times q}$; predictions use $W_b^\top \Phi_b$.
κ, λ	Decoder sensitivity budget and ridge regularization parameter.
$R(f)$, R^*	Population mean squared error of a fitted forecast and infimum over a stated decoder class.
$W_{\kappa,A}$	Population pairwise obstruction, using budget κ and conditioning variable A; distinct from coefficient matrices W_b.
T_b, $\mathcal{W}_b$, B_b^{blk}	Scalar block score for b observations, its expectation, and its deterministic range bound; $\mathcal{W}_b$ is distinct from the coefficient matrix W_b.
n, K	Number of calibration episodes and number $K = \lfloor n/2 \rfloor$ of disjoint episode pairs.
$Z_{i,jh}$, $Q_{r,jh}$	Loss of candidate j at horizon h on episode i, and obstruction score on pair r.
L_{jh}, U_{jh}	Lower obstruction endpoint and upper fitted-risk endpoint; the unindexed $U = \Gamma(X)$ instead denotes a random representation.
D_h, B_h	Diameter of the known horizon-h output set and squared-loss bound $B_h = D_h^2$.
$c(X)$, π_a, N_a, K_a	Frozen stratum label, population stratum mass, sample count, and within-stratum pair count.
W_{str}, L_{str}, $\mathcal{P}$	Stratified population witness, its lower endpoint, and the confidence set for stratum masses.
c_j, c_-, c_+, c^*	Declared class cost, lower and upper certified cost endpoints, and minimum feasible cost.
ϵ_h, δ, α	Requested RMSE tolerance, total failure probability, and failure probability allocated to one confidence statement.

Table 1: Recurring notation. Mean squared error (MSE) is an expected squared Euclidean error; root-mean-square error (RMSE) is its square root. Experiment-specific coordinate averaging is stated separately.

subspaces together with a locality term. Each construction specifies which histories receive similar coordinates. A soft association retains a weight for every group, while a hard assignment retains only the selected group. These geometric choices need not preserve all information relevant to forecasting.

For block b, partition the training histories into $C_b \geq 1$ reference groups, indexed by $c = 1, \ldots, C_b$. Let $\mathcal{X}$ be the history domain and let $z(x) \in \mathbb{R}^{d_x}$ be the preprocessed history, with dimension d_x. Each group has $J_{bc} \geq 1$ local KAHM reconstruction models, fitted using its training histories (Kumar et al., 2024, 2025; Kumar and Kargaran, 2026).

For one such local model, suppress the indices (b, c, j) and write its $N_{\mathrm{loc}} \geq 2$ reference histories as $z_1, \ldots, z_{N_{\mathrm{loc}}} \in \mathbb{R}^{d_x}$, with at least one direction of nonzero empirical variance. The fitted linear encoding $P \in \mathbb{R}^{r \times d_x}$ maps histories to $r \geq 1$ principal-component coordinates. The encoding retains only principal-component directions with positive empirical variance. Let $\Sigma \in \mathbb{R}^{r \times r}$ be the resulting nonsingular sample covariance of the encoded references. The Gaussian kernel, its reference Gram matrix, and its query vector are

$$k(\xi, \xi') = \exp\left\{-\frac{(\xi - \xi')^\top \Sigma^{-1} (\xi - \xi')}{2r}\right\},$$
$$K_{i\ell} = k(Pz_i, Pz_\ell), \qquad k_x = (k(Pz_i, Pz(x)))_{i=1}^{N_{\mathrm{loc}}}.$$

Here $\xi, \xi' \in \mathbb{R}^r$. For the local regularization parameter $\eta > 0$, kernel regression gives $h(x) = (K + \eta I_{N_{\mathrm{loc}}})^{-1} k_x \in \mathbb{R}^{N_{\mathrm{loc}}}$. Its coordinate h_i uses regression targets equal to one at reference i and zero at the other references. The reconstruction and its normalizer are

$$s(x) = \sum_{i=1}^{N_{\mathrm{loc}}} h_i(x), \qquad \widetilde{s}(x) = \begin{cases} s(x), & s(x) \neq 0, \\ 1, & s(x) = 0, \end{cases}$$
$$\widehat{z}(x) = \sum_{i=1}^{N_{\mathrm{loc}}} \frac{h_i(x)}{\widetilde{s}(x)} z_i.$$

The local regularization parameter also depends on the preprocessed values. Collect the reference histories in the matrix $Z = [z_1, \ldots, z_{N_{\mathrm{loc}}}]$ of size $d_x \times N_{\mathrm{loc}}$, and let $v_z = \|Z\|_{\mathrm{F}}^2 / (d_x N_{\mathrm{loc}})$. Starting at $\eta_0 = 5v_z/2$, the implementation computes

$$B_k = (K + \eta_k I_{N_{\mathrm{loc}}})^{-1}, \qquad \eta_{k+1} = 2v_z + \frac{\|Z - ZKB_k\|_{\mathrm{F}}^2}{d_x N_{\mathrm{loc}}}.$$

It stops when $|\eta_{k+1} - \eta_k| \leq 10^{-3}$ or after 100 iterations, and retains the last computed B_k; thus $\eta = \eta_k$ in the query map above. Nonzero empirical variation implies $v_z > 0$, so every inverse uses positive regularization. This parameter is separate from the forecast-decoder ridge parameter below. When $s(x) \neq 0$, the reconstruction coefficients sum to one and may be signed, so $\widehat{z}(x)$ is an affine combination of reference histories. The zero-sum convention defines the implementation at all queries but need not preserve this affine-combination property.

Restoring the indices, let $\widehat{z}_{bcj}(x)$ denote the reconstruction from local model j of group c in block b. Its folding score combines reconstruction error and angular discrepancy (Kumar and Kargaran, 2026, Section 3.3). Define

$$e_{bcj}(x) = 1 - \exp\{-\|z(x) - \widehat{z}_{bcj}(x)\|_2\}, \qquad \theta_{bcj}(x) = \frac{\arccos \gamma_{bcj}(x)}{\pi},$$

where $\gamma_{bcj}(x)$ is the cosine similarity $\langle z(x), \widehat{z}_{bcj}(x) \rangle / (\|z(x)\|_2 \|\widehat{z}_{bcj}(x)\|_2)$, clipped to $[-1, 1]$; when either vector has zero norm, the implementation sets this similarity to zero. The inverse cosine $\arccos$ takes values in $[0, \pi]$. The group score is

$$T_{bc}(x) = \min_{1 \leq j \leq J_{bc}} \left\{ \frac{e_{bcj}(x)^2 + \theta_{bcj}(x)^2}{2} \right\}^{1/2}.$$

Thus $T_{bc} : \mathcal{X} \to [0, 1]$ is a geometric discrepancy score: smaller values indicate a closer reconstruction match. It is not a class probability. Taking the minimum allows any local model within the group to provide the match.

Fix an exponent $\omega > 0$ and offset $\tau > 0$. The exponent controls the contrast between affinities, and the offset keeps them positive. The reported runs use $\tau = 10^{-6}$. Define

$$\Phi_{bc}(x) = \frac{(1 - T_{bc}(x) + \tau)^{\omega}}{\sum_{r=1}^{C_b}(1 - T_{br}(x) + \tau)^{\omega}}, \qquad \Phi_b(x) = (\Phi_{b1}(x), \ldots, \Phi_{bC_b}(x))^{\top}. \tag{14}$$

Then $\Phi_b(x)$ lies in the probability simplex

$$\Delta_{C_b} = \{v \in \mathbb{R}^{C_b} : v_c \geq 0 \text{ for all } c,\ \mathbf{1}_{C_b}^{\top} v = 1\}.$$

The coordinates express relative reconstruction affinity and need not be calibrated probabilities. To evaluate (14) stably, set $\ell_{bc}(x) = \omega \log(1 - T_{bc}(x) + \tau)$ and $\ell_b^{\max}(x) = \max_c \ell_{bc}(x)$, then compute

$$\Phi_{bc}(x) = \frac{\exp(\ell_{bc}(x) - \ell_b^{\max}(x))}{\sum_{r=1}^{C_b} \exp(\ell_{br}(x) - \ell_b^{\max}(x))}.$$

Subtracting the maximum leaves the normalized weights unchanged and makes at least one exponent equal to zero. Consequently, the denominator is at least one in exact arithmetic, even when all unnormalized affinities are small. This preserves relative affinities without a small-denominator replacement by uniform weights.

The forecasting method combines these reconstruction features with retained blocks, a constrained output decoder, and independent calibration. In the main experiments, histories are standardized using training means and standard deviations. Section 12 separately compares training-based range scaling and bounded raw histories. The input subsample and the maximum number of local training points are fixed by each configuration. Once fitted, the feature maps remain fixed during decoder fitting and calibration.

Why input scaling can change the predictor. Even for a fixed query and reconstruction, multiplying both vectors by a positive scalar a changes the radial term to $1 - \exp\{-a\|z - \widehat{z}\|_2\}$, while leaving their angle unchanged. Translating both vectors leaves their difference unchanged but generally changes their angle; unequal coordinate scales can change both terms. Refitting after preprocessing can additionally change the reference groups, retained principal components, and local regularization parameters. Thus a covariance-normalized kernel does not make the complete KAHM representation invariant to input scaling. Training-based scaling to $[-1, 1]$ is a model choice whose effect must be evaluated on the forecasting task.

At stage s, retain the first s blocks:

$$\Gamma_s(x) = (\Phi_1(x)^{\top}, \ldots, \Phi_s(x)^{\top})^{\top} \in \prod_{b=1}^{s} \Delta_{C_b} \subset \mathbb{R}^{p_s}, \qquad p_s = \sum_{b=1}^{s} C_b.$$

A *prefix* means these first s blocks. Their cluster partitions need not be nested: preserving earlier forecasts requires retaining the earlier coordinate functions, not their partitions. The experiments append blocks of sizes $2, 4, 8, 16$, giving prefix dimensions $2, 6, 14, 30$.

5.4 A constrained multihorizon decoder

Let (x_i, Y_i), $i = 1, \dots, N$, be the training histories and their joint targets. Form the response matrix $\mathsf{Y} \in \mathbb{R}^{N \times q}$ with row i equal to $Y_i^\top$, and the feature matrix $F = [F_1, \dots, F_s] \in \mathbb{R}^{N \times p_s}$, where row i of $F_b \in \mathbb{R}^{N \times C_b}$ is $\Phi_b(x_i)^\top$. Fit an intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s \times q}$ by solving

$$\min_{a,W} \frac{1}{N}\|\mathsf{Y} - \mathbf{1}_N a^\top - FW\|_{\mathrm{F}}^2 + \lambda\|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \le \kappa. \tag{15}$$

Here $W_b \in \mathbb{R}^{C_b \times q}$ contains the coefficient rows for block b, $\lambda \ge 0$ is the ridge parameter, and $\kappa \ge 0$ is the coefficient budget. The intercept is unpenalized. The loss sums squared errors over all target coordinates and horizons, then divides by N; it is not additionally divided by q or H.

For a known output box $\mathcal{Y} \subset \mathbb{R}^q$ containing the target, the reported prediction is

$$f_s(x) = \Pi_{\mathcal{Y}}\big(a + W^\top \Gamma_s(x)\big).$$

The objective in (15) fits the unprojected affine predictor; clipping is applied at prediction time and included in every reported loss. Throughout the paper, the same affine map is written as $a + \sum_b W_b^\top \Phi_b(x)$, with $W_b \in \mathbb{R}^{C_b \times q}$.

Centering the training matrices eliminates the intercept. Projected accelerated gradient then solves the constrained quadratic. Each projection computes the block Frobenius norms, projects that nonnegative vector onto an ℓ_1 ball of radius κ, and rescales the blocks. The recorded first-order optimality gap bounds the remaining training-objective error; a small change between iterates alone would not give that bound. Appendix A derives both the projection and the gap. The fitted-risk certificate remains valid after early stopping, provided the returned model satisfies the budget used for certification.

The ridge term regularizes fitting, whereas the block constraint defines the feasible sensitivity class. Exact block duplication leaves that class unchanged, but distributing coefficients across duplicates can reduce the squared Frobenius ridge penalty. The replication results therefore concern feasible prediction classes and their population obstruction. The duplicate-block experiment uses $\lambda = 0$ to compare these feasible classes without changing the fitting penalty.

5.5 Calibration and model selection

The candidate family includes history lengths, dictionary prefixes, geometric feature types, and baseline models. Each certificate compares a representation lower bound with a fitted-model upper bound for the same observed target, origin law, and decoder sensitivity budget. Algorithm 1 summarizes the procedure; Sections 4 and 6 give the score formulas and their guarantees.

A lower endpoint above tolerance can motivate changing the representation or budget, but does not establish that a particular change will help. Paired episode losses therefore compare old and new fitted forecasts directly. They can certify an improvement even when the representation obstruction is zero. Conversely, the lower bound can exclude a restricted class even if no tested refinement attains the desired accuracy.

In the experiments, cost is the number of retained feature coordinates, with ties resolved by history length and then validation error. This measures representation size. A deployment with unequal feature-evaluation costs can use a different cost function, provided it is fixed before calibration.

Algorithm 1: Learning and certifying KAHM forecasts

1. Specify horizons, causal origins, known target bounds, RMSE tolerances $\epsilon_h \geq 0$, total failure probability $\delta \in (0, 1)$, and a candidate-cost rule.
2. Use development data to fit geometric blocks and constrained decoders for their retained prefixes. Fit comparison models on the same observed targets.
3. Freeze all candidates, preprocessing choices, sensitivity budgets, history-based strata, costs, concentration bounds, and proposed comparisons. Allocate δ across the risk bounds, stratum-mass intervals, alternative partitions, and improvement comparisons.
4. On $n \geq 4$ independent calibration episodes, compute one mean squared loss per episode and horizon. Compute obstruction scores using disjoint episode pairs or frozen-size blocks, using the declared origin sampling rule. For stratified calibration, use one prespecified origin per independent episode (or a justified episode-level stratum), pair within strata, and estimate their masses and confidence intervals.
5. Compute simultaneous lower endpoints L_{jh} and upper endpoints U_{jh} for candidate j at horizon h.
6. Accept a fitted candidate when $U_{jh} \leq \epsilon_h^2$ for every required horizon; exclude its representation class when $L_{jh} > \epsilon_h^2$ for any horizon. Return the least costly accepted candidate, if one exists, and the cost interval between the cheapest nonexcluded class and the cheapest accepted fit.
7. Equal finite cost endpoints certify minimum feasible cost among the declared classes. Exclusion of every class certifies infeasibility within that family. Otherwise report the unresolved accuracy or cost decision explicitly.

Figure 1: Forecasting and certification for a finite family fixed before calibration. The same procedure applies to alternative feature constructions. A candidate fitted after inspecting calibration needs fresh evaluation data or a separately justified adaptive guarantee.

6 Calibration and representation refinement

6.1 Independent episodes, dependent observations within episodes

An *episode* is one complete observed trajectory, or another complete sampling unit specified by the application. Let $E_1, \ldots, E_n$ be independent and identically distributed episodes, also independent of the data used to construct and freeze the models. For horizon $h \in \mathcal{H}$, let $\mathcal{T}_h$ be a fixed, nonempty set of causal forecast origins, available in every episode and used by every compared candidate. Within an episode, observations may be arbitrarily dependent and their distribution may change with time.

Index the J candidates by $j \in \{1, \ldots, J\}$ and write $H = |\mathcal{H}|$ for the number of horizons. Candidate j has a representation Γ_j, a representation pseudometric d_j, and a fixed forecaster $f_{jh} = g_{jh} \circ \Gamma_j$ at horizon h. Its decoder g_{jh} has a known Lipschitz budget κ_{jh}. Let $X_{i,t}$ denote the history at origin t in episode i, and let $y_{i,t+h}$ denote the corresponding single-horizon observation

(distinct from the joint target Y_t in Section 5). Define

$$Z_{i,jh} = \frac{1}{|\mathcal{T}_h|} \sum_{t \in \mathcal{T}_h} \|y_{i,t+h} - f_{jh}(X_{i,t})\|_2^2, \qquad R_{jh} = \mathbb{E} Z_{i,jh}. \tag{16}$$

Thus $Z_{i,jh}$ is one average loss per episode, and R_{jh} is its population mean. Equivalently, R_{jh} is the risk for a new episode and an independently, uniformly sampled origin from $\mathcal{T}_h$. It is not a guarantee conditional on an individual history. Changing the episode weights or the origin distribution changes the risk being certified.

Assume both targets and predictions lie almost surely in a known nonempty closed convex set $\mathcal{Y}_h$. Let $D_h = \sup_{y,z \in \mathcal{Y}_h} \|y - z\|_2 < \infty$ be its diameter, and put $B_h = D_h^2$. This gives the deterministic loss bound $0 \le Z_{i,jh} \le B_h$. Prediction clipping alone does not establish this bound when the target is unbounded. Nor do training minima and maxima establish a population range.

Form $K = \lfloor n/2 \rfloor$ disjoint episode pairs, leaving one episode unused for pairing when n is odd. For $r \in \{1, \dots, K\}$ define

$$\begin{aligned} Q_{r,jh} = \frac{1}{2|\mathcal{T}_h|} \sum_{t \in \mathcal{T}_h} \Big(& \|y_{2r-1,t+h} - y_{2r,t+h}\|_2 \\ & - \kappa_{jh} d_j\big(\Gamma_j(X_{2r-1,t}), \Gamma_j(X_{2r,t})\big) \Big)_+^2. \end{aligned} \tag{17}$$

For each j, h, the variables $Z_{i,jh}$ are independent across episodes, and $Q_{r,jh}$ are independent across disjoint pairs. Terms within an individual average need not be independent. Their ranges are $[0, B_h]$ and $[0, B_h/2]$, respectively. Write $W_{jh} = \mathbb{E} Q_{r,jh}$. This is exactly the witness in (1) for the horizon-h task when A is a uniformly sampled common origin: conditional on that origin, the two histories and targets come from independent episodes.

For $N \ge 2$ independent, identically distributed scalar observations $V = (V_1, \dots, V_N)$ in a known interval of length $b \ge 0$, define the sample mean and unbiased sample variance by

$$\bar{V} = \frac{1}{N} \sum_{i=1}^N V_i, \qquad s_V^2 = \frac{1}{N-1} \sum_{i=1}^N (V_i - \bar{V})^2.$$

For a failure probability $\alpha \in (0, 1)$, use the empirical Bernstein radius

$$\mathrm{rad}_{\mathrm{EB}}(V; b, \alpha) = \sqrt{\frac{2 s_V^2 \log(2/\alpha)}{N}} + \frac{7 b \log(2/\alpha)}{3(N-1)}. \tag{18}$$

Here $\log$ is the natural logarithm. The bounded-variable inequality of Maurer and Pontil (2009, Theorem 4), also applied to the reflected observations $-V_i$, gives each of the one-sided bounds $\mathbb{E} V_1 \le \bar{V} + \mathrm{rad}_{\mathrm{EB}}$ and $\mathbb{E} V_1 \ge \bar{V} - \mathrm{rad}_{\mathrm{EB}}$ with failure probability at most α. A joint claim for both directions requires allocation to both events.

Theorem 7 (Simultaneous two-sided episode certificate) *Suppose the J candidates and H horizons are frozen using independent development data, $n \ge 4$, and all fitted decoders satisfy their stated Lipschitz budgets. Fix $\delta \in (0, 1)$ and set $\alpha = \delta/(2JH)$. With $\bar{Q}_{jh} = K^{-1} \sum_{r=1}^K Q_{r,jh}$ and $\bar{Z}_{jh} = n^{-1} \sum_{i=1}^n Z_{i,jh}$, define*

$$L_{jh} = \max\{0, \bar{Q}_{jh} - \mathrm{rad}_{\mathrm{EB}}(Q_{\cdot,jh}; B_h/2, \alpha)\}, \tag{19}$$

$$U_{jh} = \min\{B_h, \bar{Z}_{jh} + \mathrm{rad}_{\mathrm{EB}}(Z_{\cdot,jh}; B_h, \alpha)\}. \tag{20}$$

The dot denotes the full sample over episodes or pairs, so the radius uses sample size K for $Q_{\cdot,jh}$ and n for $Z_{\cdot,jh}$. With probability at least $1-\delta$, conditional on the development data, all candidate–horizon combinations simultaneously satisfy

$$L_{jh} \le W_{jh} \le R^*_{jh} \le R_{jh} \le U_{jh}. \tag{21}$$

Here R^*_{jh} is the infimum risk over all κ_{jh}-Lipschitz decoders of Γ_j, using d_j, the same horizon, the same uniform origin law, and the same raw target.

Proof Condition on the development data so that the entire candidate family is fixed. For each j, h, apply the upper empirical Bernstein bound to the n episode losses and the lower bound to the K pair scores. Since $0 \le W_{jh}$ and $R_{jh} \le B_h$, truncating the endpoints at zero and B_h preserves validity. A union bound over these $2JH$ one-sided events gives $L_{jh} \le W_{jh}$ and $R_{jh} \le U_{jh}$ jointly, with failure probability at most $2JH\alpha = \delta$.

For any admissible decoder, Theorem 1 applies to the joint law obtained by drawing a common uniform origin and two independent episodes. It bounds W_{jh} by that decoder's risk under the same origin mixture. Taking the infimum over these decoders gives $W_{jh} \le R^*_{jh}$. Finally, the fitted decoder belongs to this class, so $R^*_{jh} \le R_{jh}$. The loss and pair statistics may use the same episodes: independence between the upper and lower statistics is unnecessary for the union bound. ■

The probability $1 - \delta$ concerns repeated draws of the calibration episodes. The theorem bounds expected squared error; it does not bound every future error by $\sqrt{U_{jh}}$ or provide a prediction interval with conditional coverage. Its independent sample sizes are n and K, not the number of overlapping windows. Additional origins may reduce the variance of an episode average, but they do not create additional independent episodes. The representation, normalization, dictionary sizes, budgets, horizons, and all baselines included in the joint guarantee must be fixed before calibration. They may depend on the development data because the theorem conditions on those data.

Stratification under the same target law. A partition can concentrate pairs on distinctions that unconditional pairing rarely encounters. Condition on the development data throughout this construction, so the representation, metric, and budget are fixed. Fix a measurable map $c : \mathcal{X} \to \{1, \dots, M\}$, with $M \ge 1$, using development data only, and write $\pi_a = \Pr\{c(X) = a\}$. For each positive-mass stratum, let W_a be the witness from two independent draws conditional on $c(X) = a$. Set $W_a = 0$ when $\pi_a = 0$. The stratified witness is

$$W_{\mathrm{str}} = \sum_{a=1}^{M} \pi_a W_a \le R^*_\kappa. \tag{22}$$

This is Theorem 1 with $A = c(X)$; weighting by the population masses preserves the original target law.

To estimate it, use $n \ge 1$ independent, identically distributed observations (X_i, Y_i) from that law. For each independent trajectory, draw one origin independently from the prespecified common origin law. This produces independent observations under the episode–origin law. Pooling all dependent windows does not. The fitted-risk upper bound may still average all prescribed origins within

each episode; a union bound does not require independence between that bound and the stratified statistic. Let $N_a = \sum_i \mathbf{1}\{c(X_i) = a\}$, where $\mathbf{1}\{\cdot\}$ denotes an indicator, and set $K_a = \lfloor N_a/2 \rfloor$. Set $U_i = \Gamma(X_i)$ and write $(U_{a,k}, Y_{a,k})$, $k = 1, \ldots, N_a$, for the observations in stratum a, listed in their original sample-index order. For $r = 1, \ldots, K_a$, their disjoint pair scores are

$$Q_{a,r} = \frac{1}{2}\left(\|Y_{a,2r-1} - Y_{a,2r}\|_2 - \kappa d(U_{a,2r-1}, U_{a,2r})\right)_+^2 .$$

Assume the target set has squared diameter $B < \infty$, so $0 \le Q_{a,r} \le B/2$.

Theorem 8 (A stratified lower certificate with estimated masses) *Fix $\delta_p, \beta_1, \ldots, \beta_M \in (0,1)$ with $\delta_p + \sum_a \beta_a < 1$. If $K_a < 2$, set $\ell_a = 0$; otherwise set*

$$\ell_a = \max\{0, \bar{Q}_a - \mathrm{rad}_{\mathrm{EB}}(Q_{a,\cdot}; B/2, \beta_a)\}.$$

Here $\bar{Q}_a$ is the mean of the K_a scores. Define

$$t_p = \sqrt{\frac{\log(2M/\delta_p)}{2n}}, \qquad p_a^- = (N_a/n - t_p)_+, \qquad p_a^+ = \min\{1, N_a/n + t_p\},$$

and the confidence set, with $\Delta_M = \{p \in [0,1]^M : \sum_{a=1}^M p_a = 1\}$ the probability simplex, $\mathcal{P} = \{p \in \Delta_M : p_a^- \le p_a \le p_a^+ \text{ for every } a\}$. Then

$$L_{\mathrm{str}} := \min_{p \in \mathcal{P}} \sum_a p_a \ell_a \;\le\; W_{\mathrm{str}} \;\le\; R_\kappa^* \tag{23}$$

with probability at least $1 - \delta_p - \sum_a \beta_a$.

Proof Condition on the complete stratum-label vector. The observations in each stratum are independent draws from its conditional law; their disjoint pair scores are therefore independent and identically distributed. For every realized count with $K_a \ge 2$, the conditional failure probability of $\ell_a \le W_a$ is at most β_a. Sparse strata satisfy it deterministically. Averaging over the labels and taking a union bound gives all score inequalities with failure probability at most $\sum_a \beta_a$; there is no additional union over possible counts. For each stratum, Hoeffding's inequality gives $\Pr\{|N_a/n - \pi_a| > t_p\} \le 2\exp(-2nt_p^2) = \delta_p/M$. A union bound over the M strata therefore places $\pi = (\pi_1, \ldots, \pi_M)$ in $\mathcal{P}$ with failure probability at most δ_p. On the intersection of these events, $\min_{p \in \mathcal{P}} \sum_a p_a \ell_a \le \sum_a \pi_a \ell_a \le \sum_a \pi_a W_a$. No independence between masses and score bounds is used. ■

The set $\mathcal{P}$ is nonempty because it contains $(N_a/n)_a$. Its linear minimum is computed by starting at p^- and assigning the remaining mass, up to each upper bound, in increasing order of ℓ_a. Known population masses instead give $L_{\mathrm{str}} = \sum_a \pi_a \ell_a$ with no mass-estimation allocation. For several frozen partitions or candidate–horizon combinations, allocate failure probability over their score events and mass events before taking the largest lower endpoint. A shared partition needs only one mass event. Sparse strata contribute no positive evidence; stratification does not increase the number of independent observations.

Proposition 9 (Exactness on compatible strata) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$. Suppose each positive-mass stratum either has a constant representation $U = u_a$ or a joint law supported on two values of (U, Y). In the first case let its optimal prediction be $\mu_a = \mathbb{E}[Y \mid c(X) = a]$; in the second use the fitted values from Proposition 3, with $b = d(u_0, u_1)$ in that construction. If a single global κ-Lipschitz decoder realizes all these values, then $W_{\mathrm{str}} = R^*_\kappa$. For scalar targets on finite support, global compatibility is equivalent to the pairwise Lipschitz inequalities between all the proposed fitted values, including values from different strata.*

Proof A constant representation gives $W_a = \mathbb{E}[\|Y - \mu_a\|_2^2 \mid c(X) = a]$ by the independent-pair variance identity. A two-atom stratum has the finite-support calculation of Proposition 3. The assumed decoder attains their weighted sum, giving equality in (22). For scalar prescribed values v_i at finitely many support points u_i, define $g(u) = \min_i\{v_i + \kappa d(u, u_i)\}$. Each function in this finite minimum is κ-Lipschitz by the triangle inequality for d, and so is their minimum. At u_j, the term $i = j$ equals v_j and every other term is at least v_j by the assumed pairwise inequalities. Thus $g(u_j) = v_j$. Conversely, a global Lipschitz decoder necessarily satisfies all these pairwise inequalities. ■

For the three-atom example, the stratum $\{3\}$ has zero witness. The stratum $\{0, 1\}$ has mass 2ε, and its two atoms are conditionally equally likely, giving conditional witness $1/4$. Consequently, $W_{\mathrm{str}} = 2\varepsilon(1/4) = \varepsilon/2 = R^*_\kappa$. Stratification can thus remove a population gap, rather than merely improve concentration. It need not improve every witness: singleton strata can remove all pairwise conflicts while incompatible predictions across strata still force positive global risk. Conditional optima cannot be combined without the global compatibility condition.

Corollary 10 (Acceptance, exclusion, and inconclusive decisions) *Let $\epsilon_h \geq 0$ be a prespecified root-mean-square error (RMSE) tolerance at horizon h. On the event in* (21)*, $U_{jh} \leq \epsilon_h^2$ certifies that the fitted forecaster attains this tolerance. If $L_{jh} > \epsilon_h^2$, no decoder in the corresponding Lipschitz class attains it. When neither condition holds, the decision is inconclusive. Selecting any member of the frozen family that satisfies $U_{jh} \leq \epsilon_h^2$ at every required horizon preserves these guarantees. In particular, one may minimize a predeclared cost over the certified candidates and abstain if there are none.*

Corollary 11 (A certificate of minimum feasible cost) *Let candidate j represent a class $\mathcal{C}_j$ of multihorizon forecasts through Γ_j, containing its fitted forecast and satisfying its stated horizon-specific Lipschitz budgets. Give it a fixed cost $c_j \geq 0$. Let c^* be the minimum cost of a class containing a forecast that meets every prescribed tolerance. The minimum of an empty set is $+\infty$. For any simultaneously valid lower bounds L_{jh} on the horizon-h risk of every member of $\mathcal{C}_j$, and upper bounds U_{jh} on its fitted forecast, define*

$$\mathcal{E} = \{j : L_{jh} > \epsilon_h^2 \text{ for some } h\}, \qquad \mathcal{A} = \{j : U_{jh} \leq \epsilon_h^2 \text{ for all } h\},$$
$$c_- = \min_{j \notin \mathcal{E}} c_j, \qquad c_+ = \min_{j \in \mathcal{A}} c_j.$$

Thus $\mathcal{E}$ contains the excluded classes and $\mathcal{A}$ contains the accepted fitted candidates. The costs c_- and c_+ refer to the cheapest nonexcluded class and cheapest accepted fit, respectively. On their joint confidence event,

$$c_- \leq c^* \leq c_+. \tag{24}$$

If $c_- = c_+ < \infty$, a least-cost accepted forecast has minimum feasible cost among the specified classes. If every class is excluded, the tolerance requirements are infeasible within their union.

Proof Every accepted fitted forecast meets the tolerances. Every class containing a forecast that meets them is nonexcluded, since one violated horizon lower bound would rule out that forecast. These two inclusions give (24) and both conclusions. A joint coefficient budget can further restrict $\mathcal{C}_j$; exclusion by its larger horizon-wise Lipschitz classes remains valid. ■

If $c_- < c_+$, a cheaper nonexcluded class may still contain an accurate forecast, so selection establishes only the lowest cost among certified candidates. In the experiments, cost counts feature coordinates rather than runtime. Also, (21) implies

$$0 \le \sqrt{R_{jh}} - \sqrt{R^*_{jh}} \le \sqrt{U_{jh}} - \sqrt{L_{jh}}.$$

Thus the endpoint gap bounds the fitted decoder's RMSE suboptimality within the stated class. A large gap does not identify optimization error: the witness and either confidence correction may be loose.

For simple sample-size calculations, one may instead prespecify the Hoeffding radius $b\sqrt{\log(1/\alpha)/(2N)}$. If a fitted risk lies at least $\gamma > 0$ below a squared-error tolerance, then

$$n \ge \frac{2B_h^2 \log(1/\alpha)}{\gamma^2}$$

is sufficient for its Hoeffding upper bound to pass that tolerance with probability at least $1 - \alpha$. If the pair witness exceeds the tolerance by γ, then

$$K > \frac{B_h^2 \log(1/\alpha)}{2\gamma^2}$$

is sufficient for the corresponding lower bound to exclude the tolerance with that probability. For acceptance, the stated sample size makes the radius at most $\gamma/2$. A one-sided Hoeffding event bounds the sample mean by the population mean plus $\gamma/2$, so the upper endpoint is at most the tolerance. For exclusion, the strict sample-size inequality makes the radius strictly smaller than $\gamma/2$. On the event $\bar{Q} \ge W - \gamma/2$, where W is the population pair witness and $\bar{Q}$ is its sample mean, the lower endpoint is therefore strictly above the tolerance. Each event has probability at least $1 - \alpha$. These are sufficient sample sizes, not optimal rates. Exclusion depends on the witness margin, rather than on an unknown oracle-risk margin. Choosing the smaller of separately computed Hoeffding and empirical Bernstein bounds requires confidence allocation to both bounds; otherwise the choice must be fixed in advance.

6.2 Refinement that preserves the meaning of the norm budget

Let $\Phi_b : \mathcal{X} \to \Delta_{C_b}$, $b = 1, \ldots, s$, be fixed dictionary blocks, where $\Delta_{C_b} = \{u \in \mathbb{R}^{C_b} : u_c \ge 0,\ \sum_c u_c = 1\}$ is the probability simplex. Retain the entire stack $\Gamma_s = (\Phi_1, \ldots, \Phi_s)$. Refinement appends a block while keeping every previous coordinate function unchanged. For stacked vectors $u = (u_1, \ldots, u_s)$ and $v = (v_1, \ldots, v_s)$ define the block maximum metric $\rho_s(u, v) =$

$\max_{1\le b\le s}\|u_b - v_b\|_2$. Its induced pseudometric on histories and the coefficient norm are

$$d_s(x,x') = \rho_s(\Gamma_s(x),\Gamma_s(x')) = \max_{1\le b\le s}\|\Phi_b(x) - \Phi_b(x')\|_2, \qquad \|W\|_{2,1} = \sum_{b=1}^{s}\|W_b\|_{\mathrm{F}}. \tag{25}$$

Here $\|W\|_{2,1}$ is a sum over *blocks*, not over individual rows. As in (15), $a \in \mathbb{R}^q$ and $W_b \in \mathbb{R}^{C_b\times q}$, so the contribution of block b to a column-vector prediction is $W_b^\top \Phi_b(x)$.

Let $\mathcal{Y} \subseteq \mathbb{R}^q$ be a nonempty closed convex output set containing Y almost surely. Write $\Pi_{\mathcal{Y}}(z) = \arg\min_{y\in\mathcal{Y}}\|z-y\|_2$ for Euclidean projection; for a box, it clips each coordinate to its allowed interval. Define

$$\mathcal{F}_{s,\kappa} = \left\{ x \mapsto \Pi_{\mathcal{Y}}\left(a + \sum_{b=1}^{s} W_b^\top \Phi_b(x)\right) : a \in \mathbb{R}^q,\ \sum_{b=1}^{s}\|W_b\|_{\mathrm{F}} \le \kappa \right\}. \tag{26}$$

The intercept is unconstrained. Group norm constraints are established tools (Bach, 2008); here the block norm makes the decoder's sensitivity compatible with retained dictionary blocks.

Theorem 12 (Nested and replication-invariant refinement) *Every decoder in $\mathcal{F}_{s,\kappa}$ is κ-Lipschitz with respect to d_s. Appending a block gives $\mathcal{F}_{s,\kappa} \subseteq \mathcal{F}_{s+1,\kappa}$ and $d_{s+1} \ge d_s$ pointwise. Consequently, the infimum prediction risk in this decoder class cannot increase. Neither can the pair witness, provided the budget, target law, and conditional pairing law remain fixed. Appending an exact copy of an existing block leaves both the decoder class and the pair witness unchanged.*

Proof For the unprojected decoder, the triangle inequality and the bound $\|Av\|_2 \le \|A\|_{\mathrm{F}}\|v\|_2$ give

$$\left\|\sum_b W_b^\top\big(\Phi_b(x) - \Phi_b(x')\big)\right\|_2 \le \sum_b \|W_b\|_{\mathrm{F}}\|\Phi_b(x) - \Phi_b(x')\|_2 \le \kappa d_s(x,x').$$

Projection onto a closed convex set is nonexpansive, so this bound also holds for the projected decoder. Assigning zero coefficients to a new block embeds each old decoder in the refined class, proving inclusion and hence nonincrease of the optimal risk. Appending a block can only increase the maximum in d_s. For each paired observation, increasing that distance can only decrease the positive-part discrepancy in (1); taking expectations proves witness monotonicity.

If block $s+1$ exactly copies block b, merge its coefficients into the old block. The prediction is unchanged and $\|W_b + W_{s+1}\|_{\mathrm{F}} \le \|W_b\|_{\mathrm{F}} + \|W_{s+1}\|_{\mathrm{F}}$, so the merged coefficients remain feasible. This proves the reverse class inclusion for duplication. A duplicate also leaves the maximum distance unchanged, and therefore leaves the witness unchanged. ■

These claims concern population optima over nested feasible classes, not the risks of separately fitted finite-sample estimators. They also do not assert that every new block contains useful predictive information. Since the Euclidean diameter of each probability simplex is at most $\sqrt{2}$, $d_s \le \sqrt{2}$ regardless of the number of retained blocks. Certifying a fitted decoder at its smaller actual norm is possible, but changes the comparator class. Refinement monotonicity is asserted only at a common budget.

The choice of norm matters even when no information is added. Let two equally likely histories satisfy $\Phi(x_0) = e_1 = (1,0)^\top$ and $\Phi(x_1) = e_2 = (0,1)^\top$, with scalar targets -1 and 1. Under a

single Frobenius constraint of radius one, the largest possible forecast separation is $\sqrt{2}$, and the best affine mean squared error (MSE) is $(2-\sqrt{2})^2/4$. After duplicating the block, the concatenated coefficient vector $(-1/2, 1/2, -1/2, 1/2)$ has norm one and predicts both targets exactly. Duplication has relaxed the effective sensitivity constraint. The block-sum constraint prevents this effect while allowing genuinely new blocks to change the geometry.

Theorem 13 (A necessary amount of geometric change) *Let d_0, d_1 be measurable pseudometrics induced on a common history space by an old and a new representation. Use the same square-integrable target, conditionally independent pairing law, and sensitivity budget $\kappa > 0$. Define the random distance increase $\Delta = (d_1(X, X') - d_0(X, X'))_+$ and assume $\|\Delta\|_{L^2} < \infty$. Let W_0, W_1 be the corresponding witnesses from* (1)*, using these induced history distances. Then*

$$\sqrt{W_1} \geq \left(\sqrt{W_0} - \frac{\kappa}{\sqrt{2}} \|\Delta\|_{L^2} \right)_+ . \tag{27}$$

In particular, if a κ-Lipschitz decoder of the new representation attains RMSE at most $\epsilon \geq 0$, then

$$\|\Delta\|_{L^2} \geq \frac{\sqrt{2}}{\kappa} (\sqrt{W_0} - \epsilon)_+ . \tag{28}$$

On any event $0 \leq L_0 \leq W_0$, replacing W_0 by L_0 on the right side of (28) *is valid simultaneously for every new representation satisfying these conditions. The new representation need not have been fixed when L_0 was computed.*

Proof Set $a = \|Y - Y'\|_2$ and $v_i = (a - \kappa d_i(X, X'))_+$ for $i = 0, 1$. The positive-part function is nondecreasing and 1-Lipschitz. Consequently, $v_0 \leq v_1 + \kappa\Delta$ pointwise, and the L^2 triangle inequality gives

$$\|v_1\|_{L^2} \geq (\|v_0\|_{L^2} - \kappa\|\Delta\|_{L^2})_+ .$$

Since $\|v_i\|_{L^2} = \sqrt{2W_i}$, dividing by $\sqrt{2}$ proves (27). If the new decoder has RMSE at most ϵ, Theorem 1 gives $W_1 \leq \epsilon^2$. Combining the two inequalities and rearranging gives (28). Finally, its right side is nondecreasing in W_0. Replacing W_0 by a nonnegative lower bound therefore weakens the necessary condition, for every admissible new representation on the same event. ■

This theorem rules out a proposed repair when its population distance increase is too small. It is a necessary condition, not a sufficient refinement rule or a guarantee that an estimated distance increase is accurate. The quantity L_0 must lower-bound the old *pair witness*; an arbitrary lower bound on old oracle risk cannot be substituted. If the budget also changes from κ_0 to κ_1, the same proof gives the necessary condition

$$\left\| (\kappa_1 d_1(X, X') - \kappa_0 d_0(X, X'))_+ \right\|_{L^2} \geq \sqrt{2} (\sqrt{W_{\kappa_0,0}} - \epsilon)_+ ,$$

where $W_{\kappa_0,0}$ is the old witness at budget κ_0. This form allows the increased separation to come from a larger budget, a changed representation, or both.

6.3 When an added block can improve prediction

The preceding theorem gives a necessary geometric condition. The next calculation gives a sufficient population condition for one proposed feature block: its cross-moment with the current residual must be nonzero, and the old decoder must have unused norm budget. A positive obstruction alone implies neither condition.

Proposition 14 (Residual correlation and budget slack) *Assume $\mathbb{E}\|Y\|_2^2 < \infty$ and consider the unprojected affine predictor*

$$f_0(x) = a + \sum_{b=1}^{s} W_b^\top \Phi_b(x), \qquad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} < \kappa.$$

Let $\psi : \mathcal{X} \to \mathbb{R}^p$ be a new feature block with $\mathbb{E}\|\psi(X)\|_2^2 < \infty$. This block may take general Euclidean values; it need not be simplex-valued. The same sum of block Frobenius norms constrains its coefficients. Define

$$r = Y - f_0(X), \qquad s_0 = \kappa - \sum_b \|W_b\|_{\mathrm{F}} > 0,$$
$$C = \mathbb{E}[\psi(X) r^\top] \in \mathbb{R}^{p\times q}, \qquad G = \mathbb{E}[\psi(X)\psi(X)^\top] \in \mathbb{R}^{p\times p}.$$

Here C is the feature–residual cross-moment, G is the feature second moment matrix, and s_0 is the unused budget. If $C \neq 0$, let $a_0 = \|C\|_{\mathrm{F}}^2$ and $b_0 = \mathrm{tr}(C^\top G C)$, where tr sums the diagonal entries. Then $b_0 > 0$, and the coefficient block

$$V = \eta C \in \mathbb{R}^{p\times q}, \qquad \eta = \min\{a_0/b_0,\ s_0/\sqrt{a_0}\}$$

is feasible when appended to the old blocks. Its risk improvement is

$$R(f_0) - R(f_0 + V^\top \psi) = 2\eta a_0 - \eta^2 b_0 \geq \eta a_0 > 0. \tag{29}$$

If Y and $f_0(X)$ belong almost surely to a nonempty closed convex set $\mathcal{Y}$, projecting the new forecast onto $\mathcal{Y}$ retains at least this improvement.

Proof The old simplex-valued blocks are bounded, so $f_0(X)$ and hence the residual r have finite second moments. The same assumption on the new block makes C and G well-defined by the Cauchy–Schwarz inequality. Also, $b_0 = \mathbb{E}\|C^\top \psi(X)\|_2^2$. If $b_0 = 0$, then $C^\top \psi(X) = 0$ almost surely, implying

$$\|C\|_{\mathrm{F}}^2 = \langle C, \mathbb{E}[\psi(X) r^\top]\rangle_{\mathrm{F}} = \mathbb{E}[r^\top C^\top \psi(X)] = 0,$$

contrary to $C \neq 0$. Thus $b_0 > 0$ and $\eta > 0$.

For any coefficient matrix $V \in \mathbb{R}^{p\times q}$, expand $\|r - V^\top \psi(X)\|_2^2$ and take expectations to obtain

$$R(f_0 + V^\top \psi) = R(f_0) - 2\langle V, C\rangle_{\mathrm{F}} + \mathrm{tr}(V^\top G V).$$

Substituting $V = \eta C$ gives the equality in (29). Since $\|V\|_{\mathrm{F}} = \eta\sqrt{a_0} \leq s_0$, appending this block respects the total budget. Since $\eta \leq a_0/b_0$, the improvement is at least $\eta a_0 > 0$. Finally, projection onto a closed convex set containing Y cannot increase the new squared error. The old forecast is

already in that set, so its reference risk is unchanged. ■

The condition on the old forecast matters. An improvement between two unprojected affine predictors does not by itself imply an improvement between their separately clipped versions. The proposition also makes no claim when the old decoder exhausts its budget. Its moments are population moments: replacing them with training estimates does not establish the same population improvement. For fitted, clipped models, we therefore compare their episode losses directly.

Fix a horizon h and two forecasts before collecting $n \geq 2$ new independent evaluation episodes. Let Z_i^{old} and Z_i^{new} be their losses on episode i, averaged over the same origins as in (16). Define

$$D_i = Z_i^{\mathrm{old}} - Z_i^{\mathrm{new}} \in [-B_h, B_h], \qquad \bar{D} = \frac{1}{n}\sum_{i=1}^{n} D_i, \qquad R^{\mathrm{old}} = \mathbb{E}Z_i^{\mathrm{old}}, \quad R^{\mathrm{new}} = \mathbb{E}Z_i^{\mathrm{new}}.$$

For $\alpha \in (0,1)$, the lower empirical Bernstein bound gives, with probability at least $1-\alpha$,

$$R^{\mathrm{old}} - R^{\mathrm{new}} \geq \bar{D} - \mathrm{rad}_{\mathrm{EB}}(D; 2B_h, \alpha). \tag{30}$$

The interval length is $2B_h$, because D_i can be negative. A positive right side certifies a risk reduction for these two fitted predictors; a nonpositive value is inconclusive. For multiple fixed comparisons, allocate failure probability over the entire comparison family.

Sequential refinement can also be adaptive. At stage s, a candidate may depend on all previous results. Conditional on this full past, its evaluation episodes must be new independent draws from the specified evaluation law. Choose stage failure probabilities $\alpha_s > 0$ in advance, with $\sum_s \alpha_s \leq \delta$, and accept a change only when its paired lower bound is positive. Conditional validity at each stage and a union bound then make every accepted improvement valid on one event of probability at least $1-\delta$. This argument does not permit repeated tuning on a fixed calibration set.

Finally, a positive obstruction need not be removable by refinement. If Y is independent of X and takes values $-a$ and a with equal probability, where $a > 0$, then the minimum risk over all predictors based on X is a^2. Under unconditioned independent pairing, a constant representation has pair witness a^2; no representation of the same observed history removes this independent future noise. Conversely, a zero witness need not imply a useful predictor. For example, suppose every episode follows the same deterministic trajectory, but its target changes across forecast origins. Pairing two episodes at the same origin gives identical targets and hence a zero witness. A constant representation still cannot predict the variation across origins; its optimal risk is $\mathbb{E}\|Y - \mathbb{E}Y\|_2^2$ under the origin law. Obstruction, fitted risk, and inconclusive decisions therefore answer distinct questions and should be reported separately.

7 Population improvement versus calibration precision

The block hierarchy removes the pair witness's fixed population gap under the assumptions of Theorem 4. It does not imply that larger blocks always give a more useful certificate at a fixed sample size. We separate these questions with the exact examples in Proposition 5 and all eight finite-support laws in Table 4. The laws were examined in the earlier pairwise comparison; the new protocol freezes the complete block-size grid and fresh random streams before drawing its observations.

Table 2: Mean lower endpoints at $n = 4096$ over all 100 repetitions of each law. R^* is the minimum risk in the same scalar Lipschitz class. Columns L_b use disjoint blocks of size b; ERM uses the full sample. A single failure budget 0.05 per law and repetition covers all six methods and all three sample sizes. The hierarchy orders population witnesses, not these finite-sample endpoints.

Law	R^*	L_2	L_4	L_8	L_{16}	L_{32}	ERM
Collapsed binary	1.0000	0.9229	0.9355	0.9375	0.9311	0.9159	0.9089
Two-atom partial	0.3315	0.2776	0.2726	0.2573	0.2330	0.1995	0.2755
Three-point V	0.2222	0.1769	0.1719	0.1589	0.1393	0.1131	0.1763
Five-point zigzag	0.5400	0.3247	0.3870	0.4203	0.4158	0.3831	0.4697
Partial collapse	0.2533	0.0590	0.0964	0.1276	0.1373	0.1244	0.2044
Noisy affine	0.2500	0.0678	0.1017	0.1280	0.1352	0.1219	0.2012
Rare conflict	0.0250	0.0001	0.0004	0.0007	0.0008	0.0005	0.0116
Affine mismatch	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Protocol. For every law, 100 independent streams supply nested samples of $n \in \{256, 1024, 4096\}$ observations. For each sample, we compare disjoint blocks of sizes $b \in \{2, 4, 8, 16, 32\}$ and the full-sample empirical-minimum bound in Appendix B.4. All methods concern the same scalar 1-Lipschitz class. Block calibration uses the sharp target-range bound B_b^{blk} for $Y \in [-1, 1]$ and the KL endpoint in (13). The full-sample comparator uses loss bound four. Each of the 18 method–sample-size endpoints receives failure probability $0.05/18$. Their simultaneous statement is per law and repetition; it is not a joint claim over all 800 streams.

At size two we evaluate the pair formula. Other block scores and the full-sample comparator use feasible dual lower values. Every one of the 45,579 distinct empirical programs has primal–dual gap at most 4.01×10^{-11} in empirical-MSE units. Repeated count vectors reuse the identical optimization result. This finite-support implementation is not a timing comparison for large continuous representations. All raw categorical draws, dual certificates, and 14,400 endpoints are retained.

Results and interpretation. For partial collapse at $n = 4096$, the mean endpoint increases from 0.0590 for pairs to 0.1373 at $b = 16$, then decreases to 0.1244 at $b = 32$. The mean uncalibrated scores continue increasing, from 0.0864 to 0.2195 to 0.2370. The decrease in the endpoint therefore reflects calibration uncertainty, not a reversal of the population hierarchy. Full-sample empirical minimization remains stronger on this law, with mean endpoint 0.2044. It is also stronger on noisy affine, five-point zigzag, and rare-conflict laws. Larger blocks are an intermediate statistical and computational choice, not a uniformly superior replacement for full-sample optimization.

When the pair witness is already exact, block growth can be unnecessary. On the partially separated two-atom law, the mean endpoint decreases from 0.2776 for pairs to 0.1995 at $b = 32$. By contrast, collapsed binary noise has a constant population witness but benefits slightly from the smaller score range and variability of moderate blocks. The affine-mismatch law remains correctly uninformative for every method because its full Lipschitz optimum is zero. These different outcomes are all shown in Table 2.

There are no observed simultaneous coverage violations in the 800 law–repetition instances. This is a numerical check against known optima, not a proof or a replacement for the confidence theorem. The exact curves in Figure 2 separately show why block size must increase to capture

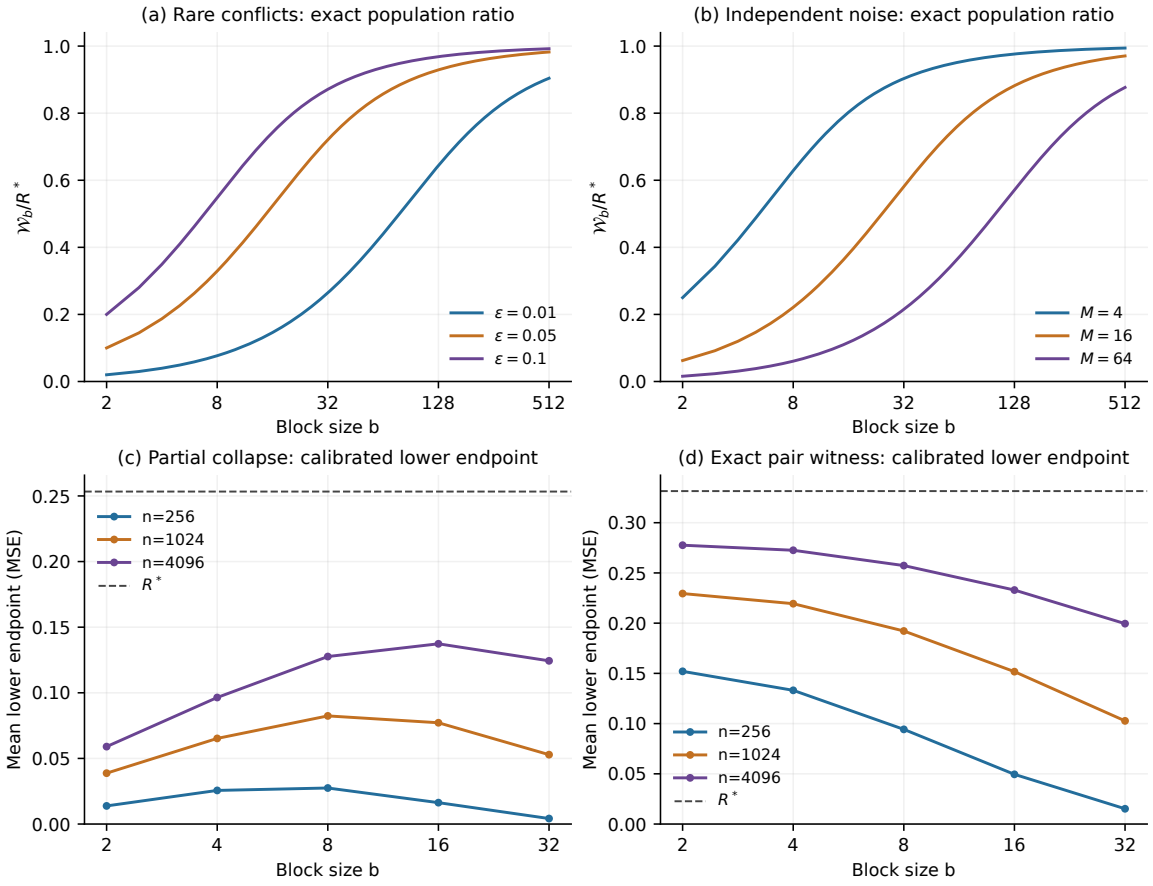


Figure 2: Population and finite-sample effects of block size. Top: exact ratios from Proposition 5, with no Monte Carlo approximation. Bottom: mean calibrated endpoints over every prescribed repetition for two contrasting laws; dashed lines are their population optima. The monotone population hierarchy does not impose monotonicity on the calibrated endpoints. Table 2 reports all eight laws, including cases favoring the full-sample comparator.

sufficiently rare conflicts or noise on many distinct inputs. Finite-sample block size should be fixed using development evidence, or selected from a family covered by a joint confidence allocation.

8 Certified decisions for learned KAHM representations

We ask whether class-level decisions can arise when the representation and the pairing partitions are learned by the actual KAHM implementation. Section 10 separately examines explicit feature maps. This is a constructed mechanism experiment with continuous histories and fresh independent episodes. It tests the connection between the theory and learned geometry; it does not estimate how often the method succeeds on natural forecasting tasks.

A smooth two-lag product and an irreducible control. Each history is $X = (X_1, X_2)$, where independently $X_j = S_j(0.9 + V_j)$, S_j is uniform on $\{-1, 1\}$, and $V_j \sim \mathrm{Unif}[-0.1, 0.1]$. The

repairable one-step target is

$$Y = 0.8X_1X_2 + E, \qquad E \sim \mathrm{Unif}[-0.05, 0.05],$$

with independent noise. This is a smooth nonlinear second-order recurrence at the declared forecast origin, with four modes in the initial-history law. In the irreducible control, an independent history $\widetilde{X}$, unavailable to the predictor, replaces X in the target formula. In both regimes $Y \in [-0.85, 0.85]$. We fix the MSE tolerance at 0.10 in these target units, after five development-only pilot seeds and before any confirmatory calibration data. It is a study-design tolerance, not an external application requirement. The noise variance is $0.05^2/3$; the irreducible control has Bayes risk $0.8^2(0.9^2 + 0.1^2/3)^2 + 0.05^2/3$. Neither reference value enters model selection or a reported confidence endpoint.

Learned classes and frozen calibration. Each of 20 independently generated trials per regime uses 2,048 training, 1,024 validation, and 16,384 calibration episodes, with one origin and one future observation per episode. Training-only coordinate min–max scaling, as defined in Section 12, precedes geometry fitting; frozen extrema transform later histories without clipping. KAHM blocks have two and four clusters, association exponent 16, retained subspace dimension 2, and at most 48 local references per cluster. The first class uses the two-coordinate block; the second retains it and appends the four-coordinate block. Thus their declared costs are 2 and 6. Both use the common sum-of-block-norms budget $\kappa = 2$ and the corresponding maximum-block distance. The cost comparison concerns these two classes and counts retained coordinates, not runtime or all possible forecasting methods.

For each class, validation selects among ridge penalties 0, 10^{-5}, and 10^{-3}. All 40 feature banks, 240 candidate fits, and 80 selected heads are recorded, and every selection is frozen before any calibration episode is generated. Twenty-two candidate fits, including nine selected fits, reach the 20,000-iteration limit before the strict 10^{-9} objective-gap criterion; their largest gaps are 5.35×10^{-9} and 2.21×10^{-9}, respectively. All coefficient budgets remain feasible. The risk guarantees require fixed feasible forecasts, not exact optimization.

Two history-only partitions take the argmax of the frozen two-cluster or four-cluster association block. No generated quadrant labels or future targets define these partitions. We compute unconditional and both stratified lower certificates, accounting for estimated stratum masses, at calibration sizes 256, 1,024, 4,096, and 16,384. A total failure allocation of 0.05 covers *both regimes, all trials, all sample-size prefixes, both classes, and all reported bounds jointly*: 0.01 for upper endpoints, 0.01 for unconditional lower endpoints, and 0.03 for the two stratified families. The latter split each partition's allocation equally between mass estimation and conditional mean bounds. All designs, sample sizes, tolerances, and partitions remain fixed after calibration starts.

What the lower information changes. At 1,024 episodes, upper calibration accepts the six-coordinate forecast in all 20 repairable trials. Stratified lower calibration additionally excludes the cheaper class in 17 trials, certifying minimum feasible cost six; the unconditional lower certificate does so in none. At 4,096 episodes both lower constructions establish this conclusion in all 20 trials. In the irreducible control, stratification excludes both classes in all 20 trials at 4,096 and 16,384 episodes, whereas unconditional lower calibration never establishes joint infeasibility at the prescribed sample sizes. Every trial at 256 episodes remains unresolved. Table 3 retains these small-sample failures.

At 16,384 episodes, the repairable six-coordinate forecast has mean empirical MSE 0.00431 and mean upper endpoint 0.00906 across trials. Its cheaper class has stratified lower endpoints between

Regime	Episodes	Upper accepts	Unconditional	Stratified
Repairable	256	0/20	0/20	0/20
Repairable	1,024	20/20	0/20	17/20
Repairable	4,096	20/20	20/20	20/20
Repairable	16,384	20/20	20/20	20/20
Irreducible	256	0/20	0/20	0/20
Irreducible	1,024	0/20	0/20	0/20
Irreducible	4,096	0/20	0/20	20/20
Irreducible	16,384	0/20	0/20	20/20

Table 3: Decisions for fitted KAHM representations over 20 fresh trials per regime. The two retained classes have costs 2 and 6, common budget $\kappa = 2$, and fixed MSE tolerance 0.10. Upper accepts counts trials in which a fitted model passes. The last two columns count certification of minimum cost 6 in the repairable regime, or exclusion of both classes in the irreducible regime. Stratification takes the maximum of separately budgeted lower bounds from two frozen learned partitions. All bounds, trials, regimes, and sample-size prefixes share a total failure allocation of 0.05.

0.2513 and 0.3905. In the irreducible control, the six-coordinate class has stratified lower endpoints between 0.3441 and 0.3657, while its unconditional lower endpoints remain between 0.0808 and 0.0929, below the tolerance. The choice of conditioning still matters: the four-cluster partition yields zero lower endpoints for the cheaper class in the repairable regime because it removes the relevant target contrast. The prespecified two-cluster partition retains that contrast. Thus stratification has decision value here, but finer conditioning is not uniformly preferable. These outcomes supply a learned-KAHM example of the class-exclusion mechanism; the natural-data results and their unresolved lower certificates retain their separate scope.

9 Lower-bound comparisons beyond two atoms

The two-atom law in Proposition 3 has an exact population witness. To distinguish a loose population bound from finite-sample uncertainty, we evaluate eight finite-support distributions (Table 4). They include deterministic and noisy targets, histories assigned identical features despite different targets, rare conflicting targets, and a case in which affine decoders are less expressive than general Lipschitz decoders. For every method, R^* is the minimum MSE over the same full scalar 1-Lipschitz class with outputs in $[-1, 1]$. These explicit representations isolate the behavior of the certificates; they do not measure the forecasting accuracy of learned KAHM features.

Protocol and comparators. The protocol was recorded before sampling. Each law uses 300 independent calibration samples at each $n \in \{64, 256, 1024, 4096\}$, totaling 9,600 trials. The four primary methods share each sample and each receive a separate one-sided failure probability 0.05; no joint 95% event across methods is asserted. We compare the disjoint-pair empirical Bernstein (EB) bound in Theorem 7, the range-aware centered bound in Appendix B.2, the dual-certified empirical risk minimization (ERM) bound with Kullback–Leibler (KL) correction in (38), and the stratified certificate in Theorem 8. The primary stratified method estimates the stratum probabilities from the

Law	Representation values u_i	Targets y_i	Stratum labels $c(u_i)$
Collapsed binary	$(0, 0)$	$(-1, 1)$	$(0, 0)$
Two atoms	$(-.6/\sqrt{2}, .6/\sqrt{2})$	$(-1, 1)$	$(0, 0)$
Three-point V	$(-1, 0, 1)$	$(1, -1, 1)$	$(0, 0, 1)$
Five-point zigzag	$(-1, -.5, 0, .5, 1)$	$(-1, 1, -1, 1, -1)$	$(0, 0, 1, 1, 1)$
Partial collapse	$(-1, -1, 0, 0, 1, 1)$	$(-1, .2, -.2, .2, -.2, 1)$	$(0, 0, 1, 1, 2, 2)$
Noisy affine	$(-1, -1, 0, 0, 1, 1)$	$(-1, 0, -.5, .5, 0, 1)$	$(0, 0, 1, 1, 2, 2)$
Rare conflict	$(3, 0, 1)$	$(0, -1, 1)$	$(0, 1, 1)$
Affine mismatch	$(-1, 0, 1)$	$(-.5, .5, -.5)$	$(0, 0, 0)$

Table 4: Prespecified joint supports for the lower-bound comparison. Entry i gives one possible pair $(U, Y) = (u_i, y_i)$. Each row assigns equal probabilities to its entries, except rare conflict, whose probabilities are $(.9, .05, .05)$. Repeated representation values retain different possible targets. Stratum labels depend only on U and are fixed before calibration. Every law uses $|u - u'|$, $\kappa = 1$, and predictions in $[-1, 1]$.

	Population quantities				Mean lower endpoint, $n = 4096$			
Law	R^*	W	W_{str}	C	Pair	Centered	ERM	Strat.
Collapsed binary	1.0000	1.0000	1.0000	1.0000	0.9306	0.9149	0.9345	0.9237
Two atoms	0.3315	0.3315	0.3315	0.3315	0.3034	0.2807	0.2908	0.3001
Three-point V	0.2222	0.2222	0.1667	0.0160	0.1996	0.0000	0.1890	0.1355
Five-point zigzag	0.5400	0.3800	0.5250	0.0744	0.3408	0.0186	0.4891	0.4442
Partial collapse	0.2533	0.0867	0.2533	0.0000	0.0646	0.0000	0.2178	0.1772
Noisy affine	0.2500	0.0972	0.2500	0.0000	0.0777	0.0000	0.2145	0.1801
Rare conflict	0.0250	0.0025	0.0250	0.0000	0.0000	0.0000	0.0148	0.0059
Affine mismatch	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 5: Finite-support comparison on the same full scalar Lipschitz class. R^* is the minimum MSE, W the unconditional pair witness, W_{str} the prespecified stratified witness, and C the centered population bound. The last four columns average 300 independent calibration samples. Each method separately has nominal one-sided confidence 95%; no joint confidence across columns is asserted. Empirical risk minimization (ERM) uses a dual-certified empirical minimum and bounded-loss Kullback–Leibler (KL) inversion. Primary stratification estimates the stratum masses. All quantities are in MSE units.

calibration data. Both primary pairwise methods use the generic score range $[0, 2]$, obtained from a target diameter of two. Prespecified secondary analyses tighten this range using the known support, use known stratum masses, or replace the empirical-minimum correction by Hoeffding's bound. The result archive includes every secondary analysis.

The empirical and population optimization programs use the full Lipschitz class, with global extension from the finite support. Feasible primal and dual values bracket each optimum; the largest empirical gap is 4.01×10^{-11}. Analytic checks cover all eight population optima. An unprojected affine optimum is never used as a Lipschitz lower bound: in the mismatch law, the former is $2/9$ and the latter is zero.

Population slack and finite-sample cost. Table 5 reports population quantities separately from mean calibrated endpoints. On partial collapse, stratification raises the population witness from 0.0867 to the exact minimum 0.2533. At $n = 4096$, its mean endpoint is 0.1772, compared with 0.0646 for unconditional pairing and 0.2178 for empirical minimization. Stratification also improves the five-point and noisy-affine witnesses. It worsens the three-point witness, from 0.2222 to 0.1667, demonstrating that a fixed partition need not help.

Empirical minimization gives the largest mean endpoint on most richer laws; the pair bound is larger on the partially separated two-atom and three-point laws at $n = 4096$. Accounting for a known representation range substantially improves centered calibration when the representation is constant. The two-atom comparison in Section 11 quantifies how treating a known zero variance as unknown can weaken a centered endpoint. Consequently, the relative performance of pairwise and centered certificates depends on the estimator and the structural information supplied to it.

Positive tolerances, coverage, and computation. The fixed RMSE tolerances are 0.10, 0.25, and 0.50 for every law. On partial collapse at $n = 4096$, unconditional pairing excludes tolerance 0.25 in 211 of 300 trials; empirical minimization and primary stratification both exclude it in all 300. Rare conflicts remain difficult: their exact risk is 0.025, but unconditional pairing and primary stratification never exclude tolerance 0.10 at this sample size. Empirical minimization excludes it in all 300 trials. Tightening the stratified score range using the known support raises its exclusion count to 284; additionally supplying known masses raises it to 300. The gain therefore depends on both pairing and available information.

All four primary methods have zero observed coverage violations across the prespecified grid. This simulation does not prove coverage; each law–size cell has only 300 repetitions, and the result archive gives binomial Monte Carlo confidence intervals. At $n = 4096$, medians across the eight law-wise median computation times are 31.3, 34.7, 265.4, and 165.0 μs for pair EB, centered, empirical minimum, and stratified EB, respectively. These single-machine measurements use a precomputed finite-support pair-score table and include sample statistics, confidence corrections, and ERM optimization after representation lookup. They exclude feature learning and do not establish a computational advantage on large continuous supports. The frozen protocol, sufficient statistics, dual certificates, and complete analysis are in `results/witness_comparison`.

10 An analytic check of decision value

An earlier controlled experiment uses explicit features to isolate the role of lower calibration; its dictionaries are not learned KAHMs. Independent episodes have history $X = (Z, S)$ and future $Y = 0.8SZ + 0.1N$ (repairable) or $Y = 0.8SR + 0.1N$ (irreducible), where $S \sim \mathrm{Bernoulli}(0.2)$ and Z, R, N are independent random signs. The retained blocks (1), $(1 - S, S)$, and $((1 + SZ)/2, (1 - SZ)/2)$ have cumulative costs $1, 3, 5$ under budget $\kappa = 2$. Their exact minimum MSEs are $0.138, 0.138, 0.01$ in the repairable regime and all 0.138 in the irreducible regime. The fixed RMSE tolerance 0.30 therefore requires the last class in one regime and is unattainable in the other. At cost 3, conditional pairing on S raises the population witness from 0.0324 to 0.138.

Each of 100 repetitions per regime uses 2048 training and 1024 validation episodes, then calibration prefixes of $256, 1024, 4096, 16384, 65536$. All decoders and the history partitions S and (S, Z) are frozen before calibration. A 0.05 failure budget covers all prefixes, sample sizes, upper endpoints, and unconditional and stratified lower endpoints within that regime–repetition.

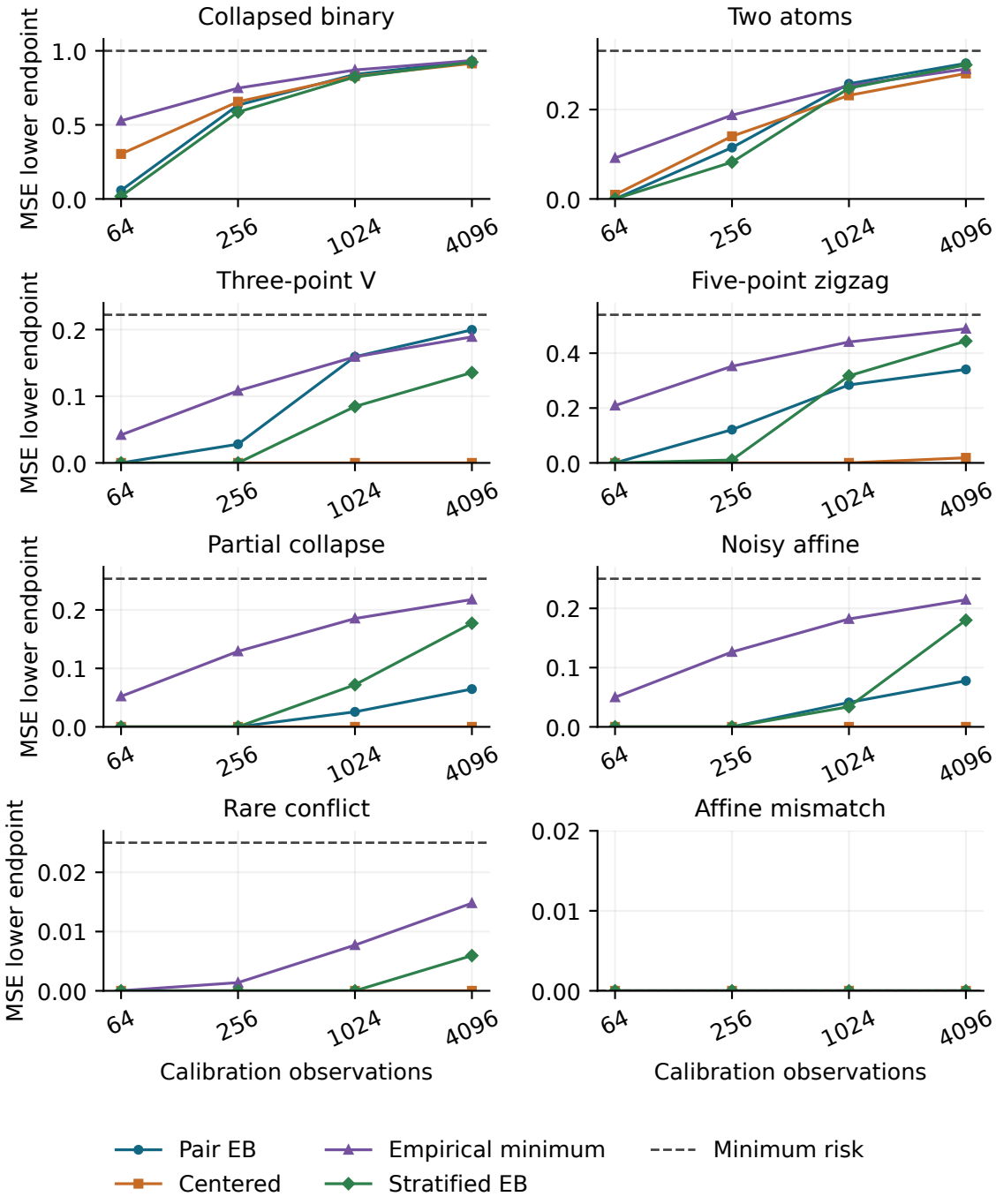


Figure 3: Mean lower endpoints over all 300 repetitions at each calibration size for every prespecified law. Dashed lines show minimum population MSE. Each method has its own 95% one-sided confidence statement. The result archive provides endpoint quantiles, Monte Carlo confidence intervals, secondary methods, and all individual trials.

Upper calibration accepts the repaired cost-5 predictor in all repetitions from $n = 1024$. Stratified lower calibration excludes both cheaper classes in all repetitions at $n = 16384$ and 65536, certifying cost 5. Unconditional lower calibration never closes the cost interval. Both rules select the same predictor; lower evidence establishes the additional class-level conclusion.

The irreducible control has no upper acceptance. Stratification excludes all classes in 1 of 100 repetitions at $n = 16384$ and all 100 at 65536; unconditional calibration never does. At $n = 4096$, unconditional pairing excludes the constant class in every repetition, while stratification excludes none. All 600 fits converge, and all 200 observed simultaneous interval events contain the exact risks. These results expose substantial sample requirements and do not establish decision frequency in natural tasks. Complete details, including all smaller-sample failures, remain in the source supplement `sections/archive/decision_value_detailed.tex` and its associated tables and figure.

11 Additional forecasting evidence

These studies test predictive accuracy and the limits of calibration beyond the controlled decision mechanisms. All prescribed configurations are retained in the result archive. Reported test RMSEs average over complete fits. Detailed study sections, tables, and fitting variability accompany the source supplement; Appendix C specifies the generators, comparisons, and independent sampling units. The matched comparison in Section 13 subsequently allows validation to select preprocessing.

11.1 Bounded synthetic episodes

Four length-72 generators cover harmonic, quasiperiodic, nonlinear-AR, and logistic observations in $[-1, 1]$. Each of five fits uses 128 training, 512 validation, 4096 calibration, and 2048 test episodes. Histories have lengths 4 or 12; horizons are 1, 4, and 12. Training-standardized geometric features and comparison models form 40 frozen candidates per fit. The 0.05 failure budget covers candidate risk and adjacent-prefix improvement bounds within each repetition; overlapping origins do not increase the independent sample size.

KAHM improves on long-horizon persistence for periodic signals, while quadratic and autoregressive regressors are stronger. On nonlinear AR, horizon-12 KAHM RMSE is 0.0743, compared with 0.0802 for ridge AR and 0.0813 for quadratic regression; these alternatives are better at one step. At all-horizon RMSE tolerance 0.35, the 800 candidates yield 301 acceptances, 24 class exclusions, and 475 unresolved decisions. No candidate certifies 0.50 on logistic data; this is not a proof of infeasibility.

Among 480 retained-KAHM candidate–horizon statements, 119 lower endpoints are positive, but only two exclude 0.35 and none excludes 0.50. Paired losses certify 72 of 360 adjacent-prefix improvements. Across all 62 main, oscillator, and budget-sensitivity fits, all 186 horizon-specific lower endpoints of the complete-family validation winners are zero; 184 of 186 are zero when selection is restricted to KAHM. Candidate exclusion and validation-winner behavior are therefore distinct findings.

11.2 Oscillators

Three new fits per system use independently initialized Van der Pol and Duffing episodes with the same sample sizes and confidence allocation. Horizon-12 KAHM sensor RMSEs are 0.3814 and

0.2499, compared with 0.2990 for Van der Pol affine-hull blocks and 0.1369 for Duffing quadratic regression. Iterated KAHM reaches 0.3408 on Van der Pol, so direct fitting also need not outperform rollout. At tolerance 0.35, the 240 candidates produce 55 acceptances, 30 exclusions, and 155 unresolved decisions. Paired KAHM refinement gains are certified in 43 of 54 Van der Pol and 10 of 54 Duffing comparisons. The risk concerns the bounded sensor under the numerical episode law, not physical-state error.

11.3 Coefficient budgets and numerical checks

Fresh streams compare $\kappa \in \{1, 4, 16\}$ on the four main systems, with three fits each and joint confidence allocation across budgets. Increasing κ from 1 to 4 helps several tasks; increasing it to 16 gives little further improvement, and conventional regressors retain advantages. Equal numerical budgets do not imply equal capacity across representations. Zero-ridge duplicate-block checks agree within solver gaps; they do not establish invariance of ridge-regularized fits.

A separate 28,800-trial two-atom check has one pair-lower violation and no centered-lower or fitted-upper violations. At complete collapse and $n = 4096$, the centered method's mean endpoint is 0.5327; using known zero representation variance gives an expected-endpoint lower bound about 0.9031, compared with mean pair endpoint 0.9177. The original comparison therefore partly reflects unused structural information, not a universal advantage over centered methods.

11.4 Original pen trajectories

PenDigits forecasting uses original, unnormalized trajectories and a fixed acquisition-box transformation. There are 1751 eligible held-out digits from 13 inferred writer identifiers. The completed-stroke eligibility rule defines a retrospective population. At horizons 1, 2, and 4, KAHM RMSEs are $0.04661, 0.06686, 0.11269$, whereas quadratic regression achieves $0.00768, 0.01777, 0.05184$; ridge AR also outperforms KAHM. Three fitting seeds reuse this writer cohort.

Even assuming iid eligible writer records, generic upper-RMSE endpoints near 1.56 certify none of $0.2, 0.35, 0.5$. Neither digits nor overlapping windows replace independent writers, and source identifiers do not establish the iid assumption. These comparisons remain descriptive.

12 Input scaling as a model choice

Training-based transformations. Section 5 explains why the complete KAHM representation is sensitive to input scaling. The forecasting experiments in Section 11 standardize each history coordinate using its training mean and standard deviation. We compare this with training min–max scaling, its clipped version, and raw histories. Here "raw" means no additional history transformation after the fixed observation transformation defining the forecasting task. For a history vector x and coordinate index j, let a_j and b_j be that coordinate's minimum and maximum over all training histories. The primary alternative is

$$z_j(x) = \begin{cases} 2(x_j - a_j)/(b_j - a_j) - 1, & b_j > a_j, \\ 0, & b_j = a_j. \end{cases} \tag{31}$$

Each nonconstant training coordinate therefore spans $[-1, 1]$. A coordinate that is constant in training is mapped to zero for every query. Extrema are fixed before geometry subsampling and never recomputed on validation, calibration, or test trajectories. Unseen values may lie outside

	Original fitting seeds				New fitting seeds			
Task	Standard	Min–max	$\Delta\%$	Wins	Standard	Min–max	$\Delta\%$	Wins
Harmonic	0.049148	0.048150	-2.03	4/5	0.050240	0.049586	-1.30	3/3
Quasiperiodic	0.037040	0.036182	-2.32	3/5	0.038290	0.035842	-6.39	3/3
Nonlinear AR	0.004561	0.004569	+0.18	4/5	0.004718	0.004656	-1.31	2/3
Logistic	0.164577	0.168931	+2.65	1/5	0.164632	0.176103	+6.97	0/3
Van der Pol	0.058343	0.060572	+3.82	0/3	0.054809	0.056049	+2.26	1/3
Duffing	0.027776	0.025812	-7.07	3/3	0.027913	0.025877	-7.29	3/3
PenDigits	0.006449	0.005173	-19.79	3/3	0.006792	0.005457	-19.65	3/3

Table 6: Validation-selected retained KAHM predictors: mean test MSE over all three horizons and fitting seeds. The primary alternative is training min–max scaling without query clipping. $\Delta\% = 100(\mathrm{MSE}_{\mathrm{minmax}}/\mathrm{MSE}_{\mathrm{standard}} - 1)$; negative values denote improvement. Wins counts fitting repetitions with a smaller all-horizon MSE. Original main tasks have five repetitions; original oscillators and PenDigits have three. All new-seed comparisons have three. Synthetic confirmation uses new development, calibration, and test episodes. PenDigits confirmation uses new training subsamples and the same held-out writer cohort, with coordinate MSE pooled before averaging horizons. Original-seed results reuse previously evaluated streams and are descriptive. No average across different task units is taken.

$[-1, 1]$; the diagnostic clipped version uses $\max\{-1, \min\{1, z_j(x)\}\}$. Clipping can map distinct out-of-range histories to the same input and is therefore a different representation, not merely a numerical safeguard. Only KAHM input geometry and models using it change in this comparison. Targets, budgets, data streams, and non-KAHM controls remain fixed. History and prefix selection use validation MSE within each preprocessing arm; test errors do not select an arm.

The first comparison reuses every original fitting seed and budget condition. Because this comparison was designed after inspecting the corresponding results, its calibration endpoints do not carry a confirmatory confidence guarantee for the subsequent choice of preprocessing. A fixed confirmation protocol uses new seeds 4701–4703 on all six synthetic systems, with fresh development, calibration, and test episodes. PenDigits instead uses new training subsamples on the same writer cohort; those comparisons remain descriptive.

Results. Table 6 reports all tasks. Mean MSE decreases on harmonic, quasiperiodic, and Duffing data in both stages; Duffing reductions are 7.07% and 7.29%. Logistic MSE increases by 2.65% and 6.97%, and Van der Pol MSE increases by 3.82% and 2.26%. The nonlinear-AR effect changes sign. Mean changes can disagree with a majority of seed-specific changes, so every paired effect and the sample standard deviation across fits are archived alongside conditional episode standard errors. Large numbers of test episodes do not replace independent fitting repetitions.

PenDigits soft-KAHM MSE decreases by 19.79% and 19.65%. Each coordinate improves at every horizon in each fit after averaging over writers. All 13 writer means improve after averaging coordinates and horizons, although horizon-four losses worsen in three original and two new-seed writer–fit comparisons. Ridge AR and quadratic regression remain more accurate. Hard KAHM assignments instead worsen by 3.96% and 4.14% in every fit. Rollout improves on 7 of seven original tasks and 6 of seven new-seed tasks, but the new-seed logistic rollout worsens by 3.61%.

Clipped and unclipped min–max results are similar; raw inputs retain the logistic and Van der Pol limitations. Budget-specific effects also vary. Neither preprocessing nor budget is selected by test performance. All horizon-specific, uncertainty, diagnostic-arm, and budget tables remain in the reproducible result archive. At tolerances $0.20, 0.35, 0.50$, primary scaling leaves the all-horizon upper-certification counts unchanged in every reported synthetic condition. The allocations cover statements within each arm, not selection across arms. Fresh synthetic calibration supports those within-arm claims. Reusing the main-study streams or changing fitting seeds on the same PenDigits cohort does not supply an independent calibration sample for the additional preprocessing choices. The results indicate that the effect of preprocessing depends on both the task and the forecasting architecture. Preprocessing can be selected using validation data before calibration, as in the matched comparison below; alternatively, selecting among calibrated arms requires a failure-probability allocation that covers all such arms.

13 A matched comparison of geometric representations

Design. To separate representation and decoder effects, we compare KAHM, RBF, and affine-hull geometry with three decoders: direct fitting under the block-norm budget in (15), direct fitting under an ordinary Frobenius-norm budget, and feature rollout. We use *head* as a short name for these decoder choices. Every geometry–head pair receives the same prefix dimensions $2, 6, 14, 30$, two history lengths, and four preprocessing options. Training, horizons, origins, regularization, and geometry subsampling follow the original study. Equal numerical budgets and feature counts do not imply equal statistical capacity or computational cost.

Five seeds, 5701–5705, generate fresh episodes for all six synthetic tasks. PenDigits uses new fitting randomness within the same development and held-out writer cohorts. Before calibration, each geometry–head pair selects preprocessing, history, and dimension by validation MSE across horizons, with ties resolved by feature count, history, preprocessing order, and name. PenDigits selects coordinates separately, then averages their losses and horizons within each writer, giving the 13 writers equal weight. Its seed variation describes repeated fitting on that cohort.

Confidence accounting. Within each synthetic task and seed, the four preprocessing arms receive 0.01 each: 0.008 for candidate risk bounds and 0.002 for refinement comparisons. The remaining 0.01 is divided among six primary geometry comparisons. For each, the alternative's loss minus KAHM's loss is averaged within each independent calibration episode over horizons. These differences lie in $[-4, 4]$; the paired empirical Bernstein bound (30) uses range length 8 and tail probability $0.01/6$. The total event has confidence at least 95% within that synthetic task–seed combination. Fixed-dimension and head-selected analyses are descriptive. PenDigits was examined in earlier studies, so its follow-up endpoints and paired writer effects provide no new confirmatory population guarantee, even under an iid-writer assumption.

Results and their limits. All 13,440 candidate fits completed. Table 7 combines absolute MSE, its variation across fits, relative changes, and paired seed signs for all 42 contrasts. KAHM has a lower mean MSE in 28 contrasts and a higher mean MSE in 14. These contrasts share models and tasks, so their signs cannot be treated as independent observations. On nonlinear AR, KAHM has lower MSE than both geometric alternatives under every head and seed. Its block-direct MSE 0.00456 also improves on ridge AR 0.00511 and quadratic regression 0.00523.

		MSE $\times 10^3$: mean (SD)			KAHM change (wins)	
Task	Head	KAHM	RBF	Affine	vs. RBF	vs. Affine
Harmonic	Block	49.77 (1.33)	50.97 (0.98)	52.45 (0.78)	-2.4% (5)	-5.1% (5)
	Frobenius	47.67 (1.31)	48.41 (1.13)	44.94 (1.13)	-1.5% (5)	+6.1% (0)
	Rollout	52.26 (1.04)	52.38 (0.98)	51.66 (1.42)	-0.2% (3)	+1.2% (1)
Quasiperiodic	Block	34.26 (1.32)	34.95 (0.70)	34.74 (2.82)	-2.0% (2)	-1.4% (2)
	Frobenius	33.72 (1.30)	34.30 (0.61)	25.92 (2.72)	-1.7% (3)	+30.1% (0)
	Rollout	39.22 (3.18)	41.75 (3.83)	37.95 (4.58)	-6.1% (2)	+3.3% (1)
Nonlinear AR	Block	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Frobenius	4.56 (0.05)	4.69 (0.04)	4.77 (0.04)	-2.7% (5)	-4.4% (5)
	Rollout	4.57 (0.08)	4.81 (0.07)	4.81 (0.05)	-5.0% (5)	-4.9% (5)
Logistic	Block	165.48 (4.28)	167.18 (2.74)	173.27 (2.22)	-1.0% (3)	-4.5% (5)
	Frobenius	160.66 (5.06)	161.29 (3.38)	170.63 (2.33)	-0.4% (2)	-5.8% (5)
	Rollout	167.63 (5.93)	164.90 (5.64)	177.74 (2.40)	+1.7% (1)	-5.7% (5)
Van der Pol	Block	52.92 (5.56)	61.82 (3.52)	48.35 (3.07)	-14.4% (5)	+9.5% (2)
	Frobenius	26.42 (4.55)	30.92 (7.17)	13.03 (2.92)	-14.5% (3)	+102.7% (0)
	Rollout	33.36 (4.97)	40.23 (16.47)	17.30 (3.63)	-17.1% (4)	+92.8% (0)
Duffing	Block	24.82 (1.60)	23.91 (2.21)	21.95 (2.32)	+3.8% (3)	+13.1% (1)
	Frobenius	23.41 (1.76)	21.21 (2.77)	13.72 (1.51)	+10.4% (0)	+70.6% (0)
	Rollout	27.41 (3.03)	27.02 (4.58)	15.23 (1.41)	+1.4% (3)	+79.9% (0)
PenDigits	Block	5.14 (0.32)	6.09 (0.22)	5.89 (1.15)	-15.6% (5)	-12.7% (4)
	Frobenius	5.04 (0.32)	6.09 (0.22)	5.14 (0.70)	-17.2% (5)	-1.9% (4)
	Rollout	5.46 (0.40)	6.65 (0.34)	5.54 (0.97)	-17.9% (5)	-1.4% (3)

Table 7: Matched geometry comparison after validation-only selection from identical preprocessing, history, and dimension options. Absolute all-horizon test MSE and its sample standard deviation across five fitting repetitions are scaled by 10^3. Relative changes use ratios of mean MSEs; negative values favor KAHM. Wins count repetitions with lower KAHM MSE out of five. PenDigits averages both coordinates and weights the same 13 writers equally. Every effect is descriptive; calibrated comparisons are reported separately.

On PenDigits, KAHM reduces MSE relative to RBF by 15.6%, 17.2%, and 17.9% across the three heads, with lower loss in all five seeds. Its reductions relative to affine geometry are 12.7%, 1.9%, and 1.4%, with four, four, and three favorable seeds. Conventional regressors remain stronger: Frobenius-direct KAHM MSE is 0.00504, compared with 0.00184 for ridge AR and 0.00102 for quadratic regression.

On Van der Pol, KAHM improves on RBF by 14.4–17.1%, but exceeds affine MSE by 9.5%, 102.7%, and 92.8% across the heads. Corresponding Duffing excesses are 13.1%, 70.6%, and 79.9%. Affine geometry has lower MSE than KAHM under Frobenius and rollout heads in every seed on both oscillators. Mean effects can also oppose most paired signs: favorable quasiperiodic block-direct means occur with lower KAHM loss in only two of five seeds.

The complete fixed-dimension curves are released as `paper/figures/matched_geometry_dimensions.pdf`, generated from all fixed-dimension records. On PenDigits, KAHM has lower mean MSE than RBF at every dimension and head; its ranking against affine

geometry changes with dimension. On nonlinear AR, KAHM has lower mean MSE than both alternatives at 14 and 30 features, whereas the ranking varies at smaller dimensions. Because histories and preprocessing are reselected, these comparisons are not tests of nesting for fixed blocks.

Only one of the 180 planned synthetic improvement comparisons is certified: Van der Pol seed 5703, KAHM rollout versus RBF rollout, with lower MSE-improvement endpoint 0.00107. All others remain inconclusive. This statement concerns two fitted predictors within the declared event; it is distinct from the descriptive mean effects and implies no population comparison on PenDigits.

14 Speech-energy forecasting with group-level calibration

This second real-data domain tests whether a larger number of documented recording groups permits useful calibration. We use the official Mini Speech Commands subset (Warden, 2018; TensorFlow, 2026), with an eligibility rule, group split, observable, and model grid specified before downloading or inspecting the audio.

Observable and sampling units. Of 8000 source clips, 7178 have the prescribed mono, 16 kHz, signed 16-bit, exactly one-second format. They belong to 1630 filename identifiers, using the source convention before `_nohash_`. All clips from an identifier remain together. A fixed random permutation assigns 244 identifiers to training, 163 to validation, 978 to calibration, and 245 to testing. These contain 1159, 677, 4256, and 1086 clips, respectively. Identifiers denote recording sessions for web volunteers; they are not biometric confirmation of distinct people. Confidence statements assume iid group-record bundles and independence from development data. Identifier separation supports the evaluation design but does not establish that population sampling assumption.

Each clip is divided into 100 nonoverlapping frames of 160 samples. For frame $t \in \{0, \dots, 99\}$ and sample index $j \in \{1, \dots, 160\}$, let x_{tj} be the signed pulse-code modulation (PCM) amplitude divided by 32768. The observed energy is

$$e_t = \frac{10 \log_{10}\Big(\max\Big\{160^{-1} \sum_{j=1}^{160} x_{tj}^2, 10^{-6}\Big\}\Big) + 60}{60} \in [0, 1].$$

The implementation clips numerical output to $[0, 1]$. A frame is observed at its end, so this transformation uses no future samples. It represents power in decibels relative to full scale, floored at -60 dB. Multiplying normalized RMSE by 60 gives RMSE in dB for this floored observable, not waveform error or error in unfloored acoustic power. The source selects and quality-screens one-second recordings; our complete-clip eligibility further restricts the population. This is retrospective forecasting within released clips, without per-utterance normalization by future means, peaks, or extrema.

Histories contain 4 or 12 frames, horizons are 1, 4, or 12 frames (10, 40, or 120 ms), and all candidates share origins $11, 15, \dots, 87$. Losses average over origins and clips within each identifier, then weight identifiers equally. Thousands of clips and windows do not replace the 978 independent calibration units assumed by the theorem.

Development and calibration. Three fitting seeds use the same group split. Each samples 4096 training histories with equal identifier probability and evaluates all validation groups with equal weights. The prespecified study compares 486 candidates per fit across eight families (Table 8). The geometric families share prefix sizes $2, 6, 14, 30$, standardization or raw inputs, budgets $1, 4$, and ridge parameters $10^{-4}, 10^{-2}$. RBF and affine affinity multipliers are $0.25, 1, 4$; KAHM uses its

Family	Test RMSE in dB: mean (SD) 10 ms	40 ms	120 ms	Maximum upper RMSE in dB 10 ms	40 ms	120 ms	All ≤ 18 dB Fits / 3
KAHM blocks	4.00 (0.15)	7.58 (0.14)	12.73 (0.06)	8.15	11.64	16.37	3/3
RBF blocks	3.79 (0.01)	7.45 (0.01)	12.72 (0.00)	7.83	11.41	16.30	3/3
Affine-hull blocks	4.28 (0.43)	7.77 (0.24)	12.81 (0.05)	8.80	11.98	16.41	3/3
Ridge AR	3.49 (0.00)	7.22 (0.01)	12.51 (0.01)	7.53	11.19	16.15	3/3
Quadratic	3.52 (0.01)	7.24 (0.03)	12.47 (0.03)	7.55	11.21	16.12	3/3
Gaussian kernel	3.51 (0.00)	7.17 (0.01)	12.39 (0.02)	7.57	11.15	16.02	3/3
Persistence	3.56 (0.00)	7.71 (0.00)	14.70 (0.00)	7.58	11.61	18.08	0/3
Constant	15.67 (0.01)	15.63 (0.01)	15.49 (0.01)	19.21	19.17	19.04	0/3

Table 8: Speech-energy forecasting on 245 held-out identifiers. Test means and sample standard deviations use three fits on the same group split. Upper columns take the maximum over those fits, with simultaneous coverage of all 72 selected family–fit–horizon statements under the stated iid group-record assumption. All six learned families pass the 18 dB requirement at every horizon in each fit; none passes the all-horizon 6 or 12 dB requirement. Errors concern the floored energy observable.

inherited reconstruction. Conventional regression heads use ridge regularization without a coefficient-norm constraint; thus their comparison with the geometric heads does not control statistical capacity. Appendix C.6 specifies their complete grids. Every family has both histories. Selection minimizes validation MSE across horizons; all 24 selected models are frozen before held-out losses are evaluated.

For each selected model and horizon, calibration gives one bounded loss in $[0, 1]$ per identifier. The bounded-mean relative-entropy upper bound in Appendix B.3 uses tail probability $0.05/72$, covering all $3 \times 8 \times 3$ statements jointly. Prespecified tolerances of 6, 12, and 18 dB are explicit resolution levels, not claimed application requirements. Model choices and calibration formulas remain fixed during held-out evaluation. Appendix C.6 describes the scoring order and its verification.

Findings. At 120 ms, mean test RMSE is 12.73 dB for KAHM, 12.51 for ridge AR, 12.47 for quadratic regression, and 12.39 for Gaussian-kernel regression. The latter three also outperform KAHM at shorter horizons. KAHM improves on long-horizon persistence, whose mean is 14.70 dB. These are descriptive accuracy comparisons; no separately calibrated superiority claim is made. The displayed standard deviations describe fitting variation on the same 245 test groups.

All six learned families satisfy the all-horizon 18 dB upper criterion in all three fits. Persistence and the constant model do not; no family passes the all-horizon 6 or 12 dB criterion. Thus the larger calibration cohort can resolve a declared tolerance under the stated sampling model, while leaving tighter requirements unresolved. The experiment supports the certification procedure without establishing that its tolerance is sufficient for an audio application or that KAHM is the best forecaster. Shared geometry costs and inference times are archived separately and imply no general computational advantage.

Input scaling on the same recording cohort. A descriptive comparison evaluates standardization, raw energy histories, training min–max scaling to $[-1, 1]$, and its clipped variant for all three geometric families. It retains the same observations, 4096 sampled training histories per seed, and decoder grids. Each transformation is fitted before the 1600-history geometry subsample. There are

	Standard	Raw	Min–max	Clipped	$\Delta\%$	Fits
KAHM	22.570 (0.197)	21.800 (0.441)	22.251 (0.092)	22.251 (0.092)	-1.41	3/3
RBF	21.466 (0.034)	21.450 (0.023)	21.304 (0.132)	21.304 (0.132)	-0.76	3/3
Affine hull	22.493 (0.665)	22.995 (0.661)	22.547 (0.104)	22.547 (0.104)	$+0.24$	1/3

Table 9: Descriptive speech scaling comparison on the same 245 test groups. Entries are all-horizon MSE means (sample standard deviations across three fits), multiplied by 10^3. Each fit averages the three horizon MSEs with equal weight. All model choices within an arm use validation data. $\Delta\%$ compares the min–max and standardized means; negative values favor min–max. "Fits" counts the paired fits with lower min–max MSE. The clipped arm shares training extrema and clips held-out inputs to $[-1, 1]$. This reused-cohort follow-up supplies no new confidence or certification claim.

2688 candidate fits in total; validation selects one model within each family, seed, and preprocessing arm. All 36 selections are frozen before test scoring. Because this analysis reuses the previously examined recording cohort, its results provide no independent confirmation or new risk certificate. Calibration groups are not scored in this comparison.

Table 9 shows that min–max scaling lowers KAHM's mean all-horizon MSE by 1.41% relative to standardization, with lower MSE in each of the three fits. Raw energy histories nevertheless give lower KAHM MSE than either min–max variant in every fit. RBF min–max MSE decreases by 0.76% with improvement in all three fits; affine-hull mean MSE increases by 0.24%, with improvement in only one fit. Clipped and unclipped results agree at the displayed precision. The ridge AR, quadratic, and Gaussian-kernel regressors from Table 8 retain lower test MSE than every KAHM scaling arm at every horizon and seed. These comparisons describe the specified implementations and the same 245 test groups.

15 Discussion and limitations

What a lower certificate adds. An upper certificate establishes that a fitted forecast meets a tolerance. A lower certificate can exclude every decoder in a specified sensitivity class. Together they can identify the minimum feasible representation cost among predefined alternatives, even when upper-only selection returns the same predictor. The learned-KAHM decision study connects this distinction to fitted geometric representations on fresh independent episodes. Its constructed mechanism isolates a recoverable forecasting signal and an independent-future control; it does not establish how often such decisions can be resolved on naturally acquired data. Costs refer to the declared feature counts, not globally optimal forecasting complexity or runtime.

Population slack and calibration precision. Proposition 2 identifies two sources of pair-witness slack: a nonzero conditional residual mean and a loose geometric inequality for paired residuals. The block hierarchy supplies a systematic route beyond the information in individual pairs. Its monotonicity concerns population expectations. Convergence to the common decoder's optimal risk requires unconditional iid draws, bounded scalar targets, and compact representation support. Conditional blocks instead approach an average of conditional oracle risks, which can be smaller than the best risk of one shared decoder. A larger block leaves fewer independent scores and costs

more to optimize. It can therefore have a weaker finite-sample endpoint despite a larger population expectation. The empirical-minimum comparator remains an important alternative; the hierarchy does not establish universal statistical or computational superiority. The block benchmark studies finite-support distributions, while the learned-KAHM decision study uses stratified pair certificates. The latter therefore tests class exclusion with learned geometry, rather than a finite-sample advantage for larger blocks on that geometry.

Stratification addresses a different issue by changing which observations are compared while preserving the marginal risk. It can recover rare obstructions, but it can also remove informative comparisons or leave too few observations in a stratum. Estimated masses add uncertainty. Neither a zero pair obstruction nor an unresolved block certificate establishes representation adequacy. Furthermore, a sharp bound for the full Lipschitz class can remain loose for a restricted affine decoder family.

What refinement guarantees. Appending feature blocks preserves earlier feasible forecasts. Exact duplication leaves the block-norm class unchanged, although it can change a ridge-regularized fit. These class properties do not force independently fitted predictors to improve. Residual correlation supplies a sufficient improvement condition when coefficient budget remains available; a separate result gives a necessary representation-distance expansion for a requested tolerance. Neither result guarantees that unsupervised clustering discovers a sufficient state. The experiments freeze their candidate families before calibration. Sequential refinement using fresh evaluation blocks remains untested.

Sampling populations and adaptation. Synthetic episodes are independently generated. PenDigits supplies only 13 eligible held-out writer identifiers, inferred from source fields; its generic endpoints are uninformative. Speech supplies documented recording identifiers and 978 calibration groups, permitting informative upper bounds under the explicit iid group-record assumption. Source identifiers and disjoint splits do not establish random population sampling. Eligibility rules based on completed recordings and source quality screening further restrict both real-data populations. Repeated fitting seeds on the same cohort measure fitting variation, not independent population replications.

Target ranges follow from the synthetic laws, the PenDigits acquisition box, or the fixed speech-energy transformation. Clipping predictions cannot bound loss for an unbounded target. The certificates do not transfer across changes in dynamics, sensors, noise, or origin laws. Original-seed scaling follow-ups reuse previously examined calibration, and their endpoints are nominal. Fresh synthetic studies use the stated independent streams; PenDigits and speech scaling follow-ups on examined cohorts remain descriptive. Confidence allocations cover each declared study, not the entire research campaign.

Accuracy and preprocessing. The matched comparisons support KAHM gains on nonlinear autoregression and against RBF geometry on the fixed PenDigits cohort. Affine-hull alternatives are stronger on the oscillators, while autoregressive, quadratic, and kernel regressors outperform KAHM on speech. Scaling to $[-1, 1]$ improves some tasks and worsens others. It must be treated as a development-selected model choice, with training extrema frozen before calibration; clipping future inputs is a distinct representation. These experiments assess the stated certificate framework and forecasting implementations. They establish neither universal KAHM superiority nor an ac-

curacy frontier against broadly tuned neural forecasters. Execution times depend on hardware and implementation, and feature count alone does not determine computational cost.

16 Conclusion

Pairwise obstructions are the first member of a hierarchy of class-wide risk lower bounds. Larger blocks reduce population slack under the stated conditions, while calibration exposes the cost of estimating those bounds from finite independent data. Conditional pairing and retained feature blocks provide complementary control over representation comparisons. Together with fitted-risk upper bounds, these tools can certify achievable accuracy, exclude inadequate classes, and resolve a declared minimum-cost decision. The learned-KAHM study demonstrates that mechanism in a controlled setting; the broader experiments identify task-dependent accuracy, scaling effects, and unresolved decisions that remain part of the method's scope.

Code, data, and research transparency

The accompanying repository provides the implementation, frozen configurations, result summaries, and figure generators in `extensions/certified_forecasting`; editable manuscript files are in its `paper` directory. The forecasting, input-scaling, matched-geometry, lower-bound comparison, block-hierarchy, learned-KAHM decision, controlled decision, speech, and speech-scaling studies have separate protocols and result directories. These provide configurations, source hashes, per-episode scores, and outcomes for every prescribed configuration. Appendix C specifies the forecasting configurations and independent sampling units. The reproduction guide is `extensions/certified_forecasting/docs/REPRODUCIBILITY.md`. Public data are identified by source and checksum; download scripts preserve attribution and license information. Appendix C.6 describes the speech study's evaluation order and its implications for interpretation.

Use of AI tools. AI tools assisted with derivation exploration, implementation, literature retrieval, and manuscript preparation.

Appendix A. Convex fitting and a numerical optimality certificate

This appendix derives the projection used to fit the decoder and a computable upper bound on its training-objective suboptimality. That optimization bound is distinct from the population prediction-risk certificate: one measures how accurately a finite training problem was solved, and the other evaluates forecasting error on new episodes.

Let N be the number of training windows and $p_s = \sum_{b=1}^{s} C_b$ the total number of retained feature coordinates. Write $F = [F_1, \dots, F_s] \in \mathbb{R}^{N \times p_s}$ for the fixed feature matrix, where $F_b \in \mathbb{R}^{N \times C_b}$ contains block b. Let $\mathsf{Y} \in \mathbb{R}^{N \times q}$ contain the corresponding stacked targets, with $q = d_y H$ for d_y output coordinates at each of H horizons. Training windows may be dependent; optimization of this finite objective requires no independence assumption on its rows.

With intercept $a \in \mathbb{R}^q$ and coefficient matrix $W \in \mathbb{R}^{p_s \times q}$, solve

$$\min_{W,a} \frac{1}{N} \|\mathsf{Y} - \mathbf{1}_N a^\top - FW\|_{\mathrm{F}}^2 + \lambda \|W\|_{\mathrm{F}}^2 \quad \text{subject to} \quad \sum_{b=1}^{s} \|W_b\|_{\mathrm{F}} \le \kappa. \tag{32}$$

Here $\mathbf{1}_N \in \mathbb{R}^N$ is the all-ones column vector, $W_b \in \mathbb{R}^{C_b \times q}$ contains the rows associated with F_b, $\lambda \geq 0$ is the ridge coefficient, and $\kappa \geq 0$ is the block-norm budget. The intercept is neither penalized nor constrained. The loss sums over output coordinates and horizons, then averages over training windows; there is no additional division by H or q. Clipping is applied after fitting. Thus (32) minimizes a convex objective before clipping, not the evaluated clipped empirical loss.

Define the row means $\bar{F} = N^{-1}\mathbf{1}_N^\top F$ and $\bar{\mathsf{Y}} = N^{-1}\mathbf{1}_N^\top \mathsf{Y}$, and the centered matrices

$$F_c = F - \mathbf{1}_N \bar{F}, \qquad \mathsf{Y}_c = \mathsf{Y} - \mathbf{1}_N \bar{\mathsf{Y}}.$$

For fixed W, minimizing over the intercept gives $a^\top = \bar{\mathsf{Y}} - \bar{F}W$. Substituting this expression eliminates the intercept. With I_{p_s} the $p_s \times p_s$ identity matrix, put

$$M = F_c^\top F_c/N + \lambda I_{p_s} \in \mathbb{R}^{p_s \times p_s}, \qquad C = F_c^\top \mathsf{Y}_c/N \in \mathbb{R}^{p_s \times q}, \qquad c = \|\mathsf{Y}_c\|_{\mathrm{F}}^2/N.$$

The reduced objective and its gradient are

$$J(W) = \operatorname{tr}(W^\top M W) - 2\langle W, C\rangle_{\mathrm{F}} + c, \qquad G(W) = \nabla J(W) = 2(MW - C).$$

The Frobenius inner product is $\langle A, B\rangle_{\mathrm{F}} = \operatorname{tr}(A^\top B)$. Since M is positive semidefinite, J is convex. Moreover,

$$\|G(W) - G(V)\|_{\mathrm{F}} \leq 2\|M\|_{\mathrm{op}}\|W - V\|_{\mathrm{F}},$$

where $\|M\|_{\mathrm{op}} = \sup_{\|v\|_2=1} \|Mv\|_2$ is the operator norm. The implementation uses projected gradient steps with acceleration and monotone restarts. The following formulas assess feasibility and optimization accuracy without requiring a convergence-rate claim for that implementation.

Proposition 15 (Projection onto the block norm ball) *Let $\mathcal{C}_\kappa = \{W : \sum_b \|W_b\|_{\mathrm{F}} \leq \kappa\}$ be the feasible coefficient set, and let $A \in \mathbb{R}^{p_s \times q}$ have the same row-block partition as W. Define $a_b = \|A_b\|_{\mathrm{F}}$ and let $\mathcal{P}_\kappa(A)$ denote the nearest matrix in $\mathcal{C}_\kappa$ in Frobenius norm. If $\sum_b a_b \leq \kappa$, then $\mathcal{P}_\kappa(A) = A$. Otherwise, for $\kappa > 0$, choose the unique threshold $\theta \in (0, \max_b a_b)$ satisfying $\sum_b (a_b - \theta)_+ = \kappa$. For $\kappa = 0$, take $\theta = \max_b a_b$. The projection is*

$$\big[\mathcal{P}_\kappa(A)\big]_b = \begin{cases} (1 - \theta/a_b)_+ A_b, & a_b > 0, \\ 0, & a_b = 0. \end{cases} \tag{33}$$

In particular, $\mathcal{P}_0(A) = 0$. For every feasible W, the objective gap satisfies

$$0 \leq J(W) - J^* \leq \mathcal{D}(W) := \langle G(W), W\rangle_{\mathrm{F}} + \kappa \max_b \|G_b(W)\|_{\mathrm{F}}, \tag{34}$$

where $J^* = \min_{V \in \mathcal{C}_\kappa} J(V)$ is also the optimum value of (32)*, and $G_b(W)$ is the row block of the gradient corresponding to W_b.*

Proof For a block B_b constrained to have Frobenius norm $v_b \geq 0$, Cauchy–Schwarz gives

$$\|B_b - A_b\|_{\mathrm{F}}^2 = v_b^2 + a_b^2 - 2\langle B_b, A_b\rangle_{\mathrm{F}} \geq (v_b - a_b)^2.$$

When $a_b > 0$, equality is attained by taking $B_b = (v_b/a_b)A_b$. When $a_b = 0$, the squared distance is v_b^2 and its optimum has $v_b = 0$. Thus matrix projection reduces to the scalar problem

$$\min_{v_1,\ldots,v_s \geq 0} \sum_{b=1}^{s} (v_b - a_b)^2 \quad \text{subject to} \quad \sum_{b=1}^{s} v_b \leq \kappa.$$

If the original norms are feasible, the solution is $v_b = a_b$. If $\kappa = 0$, the only feasible vector is zero. In the remaining case the budget constraint is active. A multiplier 2θ for that constraint reduces each coordinate minimization to $\min_{v_b \geq 0}(v_b - a_b)^2 + 2\theta v_b$, whose solution is $v_b = (a_b - \theta)_+$. The sum of these coordinates is continuous and strictly decreasing from $\sum_b a_b$ to zero as θ increases from zero to $\max_b a_b$, so the required threshold exists and is unique. Rescaling each nonzero block gives (33).

The feasible set is nonempty and compact, so continuity of J ensures that J^* is attained. For the objective gap, convexity gives, for every feasible V,

$$J(W) - J(V) \leq \langle G(W), W - V \rangle_{\mathrm{F}}.$$

Also,

$$\langle G(W), V \rangle_{\mathrm{F}} \geq -\sum_b \|G_b(W)\|_{\mathrm{F}} \|V_b\|_{\mathrm{F}} \geq -\kappa \max_b \|G_b(W)\|_{\mathrm{F}}.$$

The last value is attainable by assigning the entire budget to a block with largest gradient norm, in the direction opposite to that gradient (or by $V = 0$ if the gradient is zero). Substitute an optimal V in the convexity inequality to obtain (34). ∎

The implementation records coefficient norms, the gap $\mathcal{D}(W)$, iteration count, and whether the requested gap tolerance was reached. These are floating-point diagnostics, not interval-arithmetic guarantees. The held-out risk certificate requires the returned model to satisfy the Lipschitz budget used in calibration; it does not require an optimal training solution.

For a joint multihorizon fit, let $W_{b,h}$ be the columns of block W_b associated with the output at horizon h. Then

$$\sum_b \|W_{b,h}\|_{\mathrm{F}} \leq \sum_b \|W_b\|_{\mathrm{F}} \leq \kappa,$$

so the joint budget also bounds each horizon decoder's sensitivity. For scalar outputs, $W_{b,h}$ is the column corresponding to h in the increasingly ordered horizon list, and its Frobenius norm is its Euclidean norm. Although the block constraint is invariant to exact feature duplication, the ridge term $\lambda\|W\|_{\mathrm{F}}^2$ is not: splitting coefficients among duplicate blocks changes that penalty. The duplicate-block optimization test therefore uses $\lambda = 0$.

Appendix B. Comparison certificates and bounded-loss calibration

This appendix derives the comparison bounds used in the lower-certificate experiments. Each formula applies to independent, identically distributed calibration units and a model class fixed before calibration. The comparison examines the statistical information and computational cost of these established concentration methods relative to the pair witness.

B.1 Relation to an inverse-contrastive population bound

Let (U, Y) and (U', Y') be independent copies of a representation and its scalar target, with $\mathbb{E}Y^2 < \infty$. Use the same representation pseudometric d and sensitivity budget $\kappa > 0$ in both bounds below. Write W_κ for (1) with constant conditioning variable A. For a threshold $a > 0$, define $F = \{|Y - Y'| > a\}$, $\rho = \Pr(F)$, and $M = \mathbb{E}[\mathbf{1}_F d(U, U')^2] \leq \varepsilon < \infty$. Put $v = (|Y - Y'| - \kappa d(U, U'))_+$. Since $\mathbf{1}_F|Y - Y'| \leq \mathbf{1}_F v + \kappa \mathbf{1}_F d(U, U')$, the L^2 triangle inequality gives

$\|\mathbf{1}_F v\|_{L^2} \geq (\|\mathbf{1}_F|Y - Y'|\|_{L^2} - \kappa\sqrt{M})_+$. On F, the target difference exceeds a, so the first norm is at least $a\sqrt{\rho}$. Therefore

$$\begin{aligned} W_\kappa &\geq \tfrac{1}{2}\|\mathbf{1}_F v\|_{L^2}^2 \geq \tfrac{1}{2}(a\sqrt{\rho} - \kappa\sqrt{M})_+^2 \\ &\geq \tfrac{1}{2}(a\sqrt{\rho} - \kappa\sqrt{\varepsilon})_+^2. \end{aligned} \tag{35}$$

The inverse-contrastive-loss hypothesis of Akash et al. (2021, Lemma 2 and its appendix proof) bounds M by ε. In their continuous-target setting and positive regime $\varepsilon < a^2\rho^2/\kappa^2$, their displayed population lower bound is $(a\rho - \kappa\sqrt{\varepsilon})^2/4$; it is no larger than (35) because $\sqrt{\rho} \geq \rho$. Thus the earlier paired-residual argument already implies a prediction-risk obstruction. The positive-part discrepancy gives the stronger population bound above for the same law. A comparison of finite-sample confidence bounds additionally requires the same sampling assumptions and error probability allocations.

B.2 A range-aware centered lower bound

For this scalar comparator, let $U, Y \in \mathbb{R}$ have known interval ranges of lengths D_U, D_Y, and use $d(u, u') = |u - u'|$. From $n \geq 2$ independent, identically distributed observations (U_i, Y_i), define sample means $\bar{U} = n^{-1}\sum_i U_i$ and $\bar{Y} = n^{-1}\sum_i Y_i$, and sample standard deviations

$$s_U = \left(\frac{1}{n-1}\sum_{i=1}^n (U_i - \bar{U})^2\right)^{1/2}, \qquad s_Y = \left(\frac{1}{n-1}\sum_{i=1}^n (Y_i - \bar{Y})^2\right)^{1/2}.$$

Also let $\widehat{v}_Y = n^{-1}\sum_i (Y_i - \bar{Y})^2$, which uses denominator n rather than $n-1$. Define $c_\alpha = \sqrt{2\log(1/\alpha)/(n-1)}$. For a total failure probability $\eta \in (0,1)$, use $\alpha = \eta/3$ if $D_U > 0$ and $\alpha = \eta/2$ if $D_U = 0$, and set

$$\begin{aligned} \sigma_Y^- &= \max\left\{(s_Y - D_Y c_\alpha)_+, \sqrt{\left(\widehat{v}_Y - D_Y^2\sqrt{\frac{\log(1/\alpha)}{2n}}\right)_+}\right\}, \\ \sigma_U^+ &= \begin{cases} \min\{s_U + D_U c_\alpha, D_U/2\}, & D_U > 0, \\ 0, & D_U = 0, \end{cases} \qquad L_{\mathrm{ctr}} = (\sigma_Y^- - \kappa\sigma_U^+)_+^2. \end{aligned} \tag{36}$$

Then $L_{\mathrm{ctr}} \leq R_\kappa^*$ with probability at least $1 - \eta$, where R_κ^* is the infimum risk over scalar κ-Lipschitz functions of U. Both target lower bounds hold on the same confidence event, which permits taking their maximum.

To verify this, rescale Maurer and Pontil (2009, Theorem 10) to obtain the one-sided events $\sqrt{\mathrm{Var}(Y)} \geq s_Y - D_Y c_\alpha$ and $\sqrt{\mathrm{Var}(U)} \leq s_U + D_U c_\alpha$, each with probability at least $1 - \alpha$. Also, $\widehat{v}_Y = \inf_a n^{-1}\sum_i (Y_i - a)^2$, where a may be restricted to the known target interval. Replacing one observation changes this infimum by at most D_Y^2/n, and $\mathbb{E}\widehat{v}_Y = (1 - 1/n)\,\mathrm{Var}(Y) \leq \mathrm{Var}(Y)$. For $D_Y > 0$, the bounded-differences inequality consequently gives

$$\Pr\left\{\widehat{v}_Y - \mathbb{E}\widehat{v}_Y > D_Y^2\sqrt{\frac{\log(1/\alpha)}{2n}}\right\} \leq \alpha.$$

This proves the second target lower bound; for $D_Y = 0$ it is identically zero. The deterministic interval bound $\mathrm{Var}(U) \leq D_U^2/4$ follows by bounding the mean squared distance to the interval

midpoint, and justifies the cap. A union bound combines the two target statements and, when $D_U > 0$, the representation statement. On their common event, (3) with constant A gives $R^*_\kappa \geq (\sqrt{\mathrm{Var}(Y)} - \kappa\sqrt{\mathrm{Var}(U)})^2_+ \geq L_{\mathrm{ctr}}$. When the representation is structurally constant, its variance is exactly zero and requires no confidence correction. A zero sample variance alone does not justify that substitution.

B.3 One-sided relative-entropy bounds

For probabilities $r, p \in [0, 1]$, define binary relative entropy

$$\mathrm{kl}(r\|p) = r\log(r/p) + (1-r)\log\{(1-r)/(1-p)\}.$$

Terms with zero numerator are zero. At $p = 0$ or $p = 1$, the value is infinite unless $r = p$, in which case it is zero. For $c > 0$, let $\mathrm{kl}_-(r, c)$ and $\mathrm{kl}_+(r, c)$ be the smallest and largest $p \in [0, 1]$ satisfying $\mathrm{kl}(r\|p) \leq c$. These endpoints lie respectively below and above r. They are nondecreasing functions of r and can be found by scalar bisection. In particular, $\mathrm{kl}_-(0, c) = 0$ and $\mathrm{kl}_+(1, c) = 1$.

Let $V_1, \ldots, V_n$ be independent and identically distributed in $[0, B]$, with $B > 0$, mean μ, and sample mean $\bar{V}$. For $\eta \in (0, 1)$ and $c = \log(1/\eta)/n$, each of

$$B\,\mathrm{kl}_-(\bar{V}/B, c) \leq \mu, \qquad \mu \leq B\,\mathrm{kl}_+(\bar{V}/B, c) \tag{37}$$

has failure probability at most η. A joint two-sided claim allocates failure probability to both directions. If $B = 0$, both endpoints are zero. The scores need not be Bernoulli observations.

For completeness, convexity of the exponential gives, for every real t,

$$\mathbb{E}e^{tV_1/B} \leq 1 - \mu/B + (\mu/B)e^t.$$

Write $p = \mu/B$. Independence and the exponential Markov inequality give, for $z \in (p, 1]$,

$$\Pr\{\bar{V}/B \geq z\} \leq \inf_{t \geq 0} e^{-tnz}(1 - p + pe^t)^n = e^{-n\,\mathrm{kl}(z\|p)}.$$

For $z \in [0, p)$, the same calculation with $t \leq 0$ bounds $\Pr\{\bar{V}/B \leq z\}$ by the same expression (Hoeffding, 1963). If $p \in \{0, 1\}$, boundedness makes every score constant and the assertions hold directly. For $0 < p < 1$, the divergence increases as z moves away from p on either side. Inverting these two one-sided bounds at probability η proves (37). Applying the upper endpoint to losses of a frozen forecaster gives an alternative to (20), under the same sampling and range assumptions. An episode average is one bounded score. Choosing between different concentration methods after seeing calibration requires a joint allocation, just as in Section 3.

B.4 A certified empirical-minimum comparator

An empirical-minimum lower bound provides a comparison with structural certificates that avoid optimization on calibration observations (Kumar and Kargaran, 2026, Section 2.5). Let $\mathcal{G}$ be a fixed nonempty class with losses $\ell_g(X, Y) \in [0, B]$, $B > 0$, and define the following quantities from independent, identically distributed observations (X_i, Y_i) of the target law:

$$R^* = \inf_{g \in \mathcal{G}} \mathbb{E}\ell_g(X, Y), \qquad a_n = \inf_{g \in \mathcal{G}} \frac{1}{n}\sum_i \ell_g(X_i, Y_i).$$

Assume a_n is measurable. Let $\eta \in (0,1)$ be the allocated failure probability. If a computed measurable value d_n satisfies $d_n \le a_n$ for every calibration sample, put $q_n = \max\{0, d_n/B\}$. Since $0 \le a_n \le B$, one has $0 \le q_n \le a_n/B \le 1$. Then

$$L_{\mathrm{ERM}} = B\,\mathrm{kl}_-\left(q_n, \frac{\log(1/\eta)}{n}\right) \le R^* \tag{38}$$

with probability at least $1-\eta$.

Indeed, $a_n \le n^{-1}\sum_i \ell_g(X_i, Y_i)$ for every fixed g. For $t \ge 0$, the preceding moment-generating-function bound therefore gives

$$\mathbb{E}e^{tna_n/B} \le \inf_{g\in\mathcal{G}}\{1 - R(g)/B + (R(g)/B)e^t\}^n = \{1 - R^*/B + (R^*/B)e^t\}^n,$$

where $R(g) = \mathbb{E}\ell_g(X,Y)$. The equality follows because $1 - r + re^t$ is nondecreasing in r for $t \ge 0$ and the class has predictors with risks arbitrarily close to R^*. If $R^* = 0$, this moment-generating-function bound and $a_n \ge 0$ imply $a_n = 0$ almost surely. If $R^* = B$, the conclusion follows from $L_{\mathrm{ERM}} \le B$. For $0 < R^* < B$, the exponential Markov argument yields $\Pr\{a_n/B \ge z\} \le \exp\{-n\,\mathrm{kl}(z\|R^*/B)\}$ for $z \in (R^*/B, 1]$. Its inversion bounds $B\,\mathrm{kl}_-(a_n/B, \log(1/\eta)/n)$ from above by R^* with probability at least $1-\eta$. Monotonicity in the first argument and $q_n \le a_n/B$ prove (38). Neither attainment of the population infimum nor a uniform complexity penalty is needed: the empirical infimum lies below the empirical risk of each fixed population competitor. The result lower-bounds the risk of every class member; it supplies neither an upper risk guarantee nor an excess-risk bound for a predictor fitted to these calibration observations.

For scalar Lipschitz regression on a finite calibration sample, the empirical infimum is a convex quadratic program (Gottlieb et al., 2017, Section III-A). Restricting predictions to the known target interval $[y_-, y_+]$ preserves the optimum, because clipping reduces loss and preserves the Lipschitz budget. Here $y_- \le y_+$ are known bounds with $Y \in [y_-, y_+]$ almost surely. Let $u_1, \ldots, u_{m_U}$ be the $m_U \ge 1$ distinct observed representation values and let $v_j \in [y_-, y_+]$ be the fitted value at u_j. Set

$$w_j = \frac{\#\{i : U_i = u_j\}}{n}, \qquad t_j = \frac{1}{n}\sum_{i:U_i=u_j} Y_i, \qquad \nu = \frac{1}{n}\sum_i Y_i^2.$$

Here $\#$ counts sample indices, $j = 1, \ldots, m_U$, and $U_i = \Gamma(X_i)$. Expanding the squared losses shows that the empirical risk is $\nu + \sum_j (w_j v_j^2 - 2t_j v_j)$. Minimize this expression subject to $v_j - v_k \le \kappa d(u_j, u_k)$ for every ordered pair (j,k) and $y_- \le v_j \le y_+$. These constraints are necessary for a global Lipschitz decoder. They are also sufficient: the function $g(u) = \min_j\{v_j + \kappa d(u, u_j)\}$ is measurable, κ-Lipschitz, and equals v_j at every observed node. Projecting g onto $[y_-, y_+]$ preserves those values and its Lipschitz constant. For ordered scalar inputs with absolute distance, constraints in both directions between adjacent nodes imply every pair constraint by summation.

Let $v = (v_1, \ldots, v_{m_U})^\top$ and write the L_{con} pairwise and interval constraints as $Av \le b$, with $A \in \mathbb{R}^{L_{\mathrm{con}} \times m_U}$ and $b \in \mathbb{R}^{L_{\mathrm{con}}}$. For any nonnegative multiplier vector $\zeta \in [0,\infty)^{L_{\mathrm{con}}}$, a dual lower value is

$$d_{\mathrm{dual}}(\zeta) = \nu - \zeta^\top b - \sum_{j=1}^{m_U} \frac{((A^\top\zeta)_j - 2t_j)^2}{4w_j}. \tag{39}$$

Indeed, adding $\zeta^\top(Av - b)$ to the quadratic objective gives its Lagrangian. Completing the square in each v_j shows that the unconstrained minimum occurs at $v_j = (2t_j - (A^\top\zeta)_j)/(2w_j)$ and equals

(39). Every observed node has positive frequency, so $w_j > 0$ and these divisions are defined. For feasible v, the added term is nonpositive. Hence weak duality gives $d_{\mathrm{dual}}(\zeta) \le a_n$, supplying the required d_n. A primal objective alone is an upper bound on a_n and cannot replace this dual value. The implementation records primal feasibility and the primal–dual gap; its floating-point checks do not constitute interval-arithmetic proofs.

Appendix C. Experimental protocols and comparison models

C.1 Episode laws

Each row of a generated data array is one independent episode. In the following definitions, t is an integer time index, y_t is the observed scalar signal, and $\mathrm{Unif}[a, b]$ denotes the uniform distribution on $[a, b]$. All random draws are independent unless an episode-level parameter is explicitly shared across times. The observation range is contained in $[-1, 1]$ for all four main systems. Clipping forecasts to this interval gives scalar squared losses in $[0, 4]$.

1. **Harmonic.** Draw one phase $\theta \sim \mathrm{Unif}[0, 2\pi]$ and one angular frequency $\nu \sim \mathrm{Unif}[.12, .28]$ per episode. Define
$$y_t = .85\cos(\theta + \nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03],$$
where η_t is observation noise. Thus $|y_t| \le .88$.

2. **Quasiperiodic.** Draw phases θ_1 and θ_2 independently from $\mathrm{Unif}[0, 2\pi]$, and draw $\nu \sim \mathrm{Unif}[.10, .22]$ once per episode. Set
$$y_t = .55\cos(\theta_1 + \nu t) + .30\cos(\theta_2 + \sqrt{2}\nu t) + \eta_t, \qquad \eta_t \sim \mathrm{Unif}[-.03, .03].$$
The two angular frequencies are ν and $\sqrt{2}\nu$. Again $|y_t| \le .88$.

3. **Nonlinear autoregression.** Initialize y_0 and y_1 independently from $\mathrm{Unif}[-.8, .8]$. For $t \ge 1$, apply
$$y_{t+1} = \tanh\{1.25y_t - .65y_{t-1} + .25\sin(3y_t) + \eta_{t+1}\}, \qquad \eta_{t+1} \sim \mathrm{Unif}[-.10, .10],$$
where η_{t+1} is an innovation and $\tanh$ denotes the hyperbolic tangent. The implementation discards the first 80 stored values before retaining a length-72 episode. The two initial values occupy indices zero and one; recursive updates start at index two. The hyperbolic tangent ensures the stated output bound.

4. **Logistic.** Draw the episode parameter $r \sim \mathrm{Unif}[3.65, 3.95]$ and initial internal state $u_0 \sim \mathrm{Unif}[.05, .95]$, then iterate $u_{t+1} = ru_t(1 - u_t)$. After 40 burn-in updates, store the next 72 updated values, transformed to the observed scale by $y = 2u - 1$. Because $r \le 4$, updates preserve $u \in [0, 1]$ and hence $y \in [-1, 1]$. Within-episode dynamics can be chaotic. Independence between episodes follows from independent initialization, without requiring a mixing argument.

The finite generation and origin protocol defines the risk being estimated. No assumption of stationarity after burn-in or sampling from an invariant distribution is needed. Each generator has its own observation scale and horizon interpretation; we do not average errors across generators as if they were a common physical measure.

C.2 Oscillator protocol

For continuous time t, position $x(t)$ and velocity $v(t)$ satisfy

$$\dot{x} = v, \qquad \dot{v} = (1 - x^2)v - x \qquad \text{(Van der Pol)},$$
$$\dot{x} = v, \qquad \dot{v} = -0.2v + x - x^3 + 0.3\cos(1.2t + \varphi) \qquad \text{(Duffing)},$$

where dots denote time derivatives. Initial position and velocity are independent $\mathrm{Unif}[-2, 2]$ draws; Duffing phase $\varphi \sim \mathrm{Unif}[0, 2\pi]$ is also independent per episode. Fourth-order Runge–Kutta uses step 0.025 and discards ten time units. The 72 sensor observations are

$$y_j = 0.95\tanh(x(10 + 0.1j)) + \eta_j, \qquad \eta_j \overset{\text{iid}}{\sim} \mathrm{Unif}[-0.02, 0.02], \quad j = 0, \ldots, 71.$$

Their range $[-0.97, 0.97]$ lies within the certificate range $[-1, 1]$. A prespecified half-step check on 16 fixed initializations, with phases and noise shared between steps 0.025 and 0.0125, gives maximum sensor changes 1.69×10^{-6} (Van der Pol) and 3.49×10^{-6} (Duffing), below 10^{-3}. This checks numerical consistency without a rigorous continuous-time integration-error bound.

C.3 PenDigits acquisition and development

The original UNIPEN files contain 7,494 development and 3,498 test digits, with nominal 100 ms sampling. Every coordinate is checked against the acquisition range $[0, 500]$; no stroke centering, alignment, or resampling is applied, and pen-up gaps are excluded. Writer identifiers include the source split; approximately 250 records per identifier support the inference described in Section 11.4.

The first 24 development writers supply training and the remaining six validation. Each of three seeds selects 20 eligible strokes per training writer (480 total); validation uses all 717 eligible strokes. Histories have lengths $m \in \{4, 8\}$, common zero-based origins $7, 11, 15, 19, 23, 27$, and horizons $1, 2, 4$. The 40-candidate family follows the main comparison design, with successive block sizes $2, 4, 8, 16$, $\kappa = 4$, ridge $\lambda = 10^{-4}$, and affinity exponent $\omega = 16$. Validation weights writers equally and averages MSE over horizons. Normalized RMSE 0.01 equals 2.5 acquisition pixels. For the additional iid-writer calculation, two coordinates, three fits, 40 candidates, and three horizons give 720 one-sided statements. With 13 writers, the Hoeffding addition is $4\sqrt{\log(720/0.05)/(2 \cdot 13)} = 2.42741$.

C.4 Random streams and configuration

Main fitting seeds are $1701, \ldots, 1705$. For seed s, separate NumPy streams use $100s+11$, $100s+23$, $100s + 47$, and $100s + 89$ for training, validation, calibration, and testing. Clustering and retained blocks receive deterministic offsets from the training seed and history length. Archived configurations specify these offsets and random-generator versions. Geometry uses at most 1600 training rows; decoders use all 1664 rows (128 episodes, 13 common origins). Overlapping training rows are permitted. Validation and test losses weight episodes equally and origins equally within episodes; selection averages MSE over the three horizons.

C.5 Comparison definitions

In the main synthetic campaigns, RBF and affine-hull comparators use the same type of k-means partition, with five initializations and at most 150 iterations. Here k is the number of coordinates

in the block. To define their scores, suppress the block index. Let $\xi \in \mathbb{R}^{d_x}$ be a preprocessed history, where d_x is its input dimension. For cluster $c \in \{1, \dots, k\}$, let $\mu_c \in \mathbb{R}^{d_x}$ be its fitted center, $\mathcal{I}_c$ the nonempty set of assigned geometry-training indices, and $n_c = |\mathcal{I}_c|$. Define the cluster's squared-radius scale and the global RBF scale by

$$s_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|\xi_i - \mu_c\|_2^2,\ 10^{-6}\right\}, \qquad \beta = \mathrm{median}_{1\le c\le k}\, s_c,$$

where median is the usual sample median. The RBF score is

$$a_c^{\mathrm{RBF}}(\xi) = \frac{\|\xi - \mu_c\|_2^2}{2\beta}.$$

Thus every cluster uses the same bandwidth, while the positive floor prevents division by a zero radius.

For the affine-hull comparator, take the singular value decomposition of the matrix whose rows are $(\xi_i - \mu_c)^\top$, $i \in \mathcal{I}_c$. When $n_c \ge 2$, let V_c contain its first $k_c = \min\{2, d_x - 1, n_c\}$ right singular vectors as columns. For a singleton cluster, set $k_c = 0$ and use an empty basis. Then $P_c = V_c V_c^\top$ is the orthogonal projection onto the selected local subspace; $P_c = 0$ for an empty basis. The constraint $k_c \le d_x - 1$ leaves an orthogonal direction whenever one is available. Define

$$r_c = \max\left\{\frac{1}{n_c}\sum_{i\in\mathcal{I}_c}\|(I_{d_x} - P_c)(\xi_i - \mu_c)\|_2^2,\ 0.05 s_c,\ 10^{-6}\right\},$$
$$a_c^{\mathrm{AH}}(\xi) = \frac{\|(I_{d_x} - P_c)(\xi - \mu_c)\|_2^2}{r_c} + 0.1\frac{\|\xi - \mu_c\|_2^2}{s_c},$$

where I_{d_x} is the identity matrix. The first term measures distance from the local affine subspace; the second penalizes distance from its center. The floors in r_c keep the normalization positive even when the training residuals vanish.

For either choice $a_c = a_c^{\mathrm{RBF}}$ or $a_c = a_c^{\mathrm{AH}}$, the block coordinates are

$$\Phi_c(\xi) = \frac{\exp(-a_c(\xi))}{\sum_{j=1}^k \exp(-a_j(\xi))}.$$

Normalization is evaluated in log space. The score multiplier is one in the original synthetic campaigns; subsequent studies specify their expanded grids. Both alternatives use the retained-block decoder.

Hard KAHM states replace the final block by a one-hot vector: its largest coordinate becomes one and all others become zero. Ridge AR uses the original causal history. Quadratic regression uses training-standardized histories and all monomials of degree one or two, including cross-products; it does not add a constant feature because the intercept is unpenalized. Both direct decoders use the stated Frobenius constraint and ridge coefficient. Persistence repeats the latest observation at every horizon. The constant model predicts the training target mean separately for each horizon and has sensitivity budget zero.

For the iterated baselines, let $z_t \in \mathbb{R}^{1\times p}$ be the row vector of p features at time t. Ridge regression fits an affine feature transition and a scalar observation decoder,

$$z_{t+1} \approx z_t M + b, \qquad y_t \approx z_t v + c,$$

where $M \in \mathbb{R}^{p\times p}$ is the transition matrix, $b \in \mathbb{R}^{1\times p}$ is its intercept, $v \in \mathbb{R}^p$ is the observation coefficient vector, and $c \in \mathbb{R}$ is the observation intercept. For a positive integer horizon h, composing these fitted maps gives the un-clipped forecast

$$\widetilde{y}_{t+h} = z_t M^h v + b\left(\sum_{\ell=0}^{h-1} M^\ell\right) v + c,$$

with M^0 the identity matrix. Only this final scalar forecast is clipped. The effective output coefficient is $M^h v$; its Euclidean norm is the frozen Lipschitz budget relative to Euclidean feature distance at horizon h. The certificate evaluates this fitted forecast directly and requires no spectral-stability assumption, exact feature closure, or approximation to the trajectory distribution.

C.6 Speech model grid and evaluation order

The prespecified speech study uses fitting seeds $8601, 8602, 8603$ and history lengths $m \in \{4, 12\}$. Each of the KAHM, RBF, and affine-hull families uses the four retained prefixes, two preprocessing choices (standardization and raw inputs), budgets $\kappa \in \{1, 4\}$, and ridge coefficients $\lambda \in \{10^{-4}, 10^{-2}\}$. For RBF and affine-hull features, an additional multiplier $\rho \in \{0.25, 1, 4\}$ changes the association weights to values proportional to $\exp(-\rho a_c(\xi))$, with a_c defined above. KAHM retains affinity exponent 16, at most 1600 geometry-training histories, and at most 24 reference points per local model. The resulting candidate counts are 64 for KAHM and 192 for each geometric comparator.

Linear and quadratic regression use training-standardized histories, with ridge coefficients λ in $\{10^{-6}, 10^{-4}, 10^{-2}, 10^{-1}\}$. Gaussian-kernel regression uses the same standardization and a 128-landmark Nyström approximation to $k(z, z') = \exp(-\gamma\|z - z'\|_2^2)$. Its kernel parameter is $\gamma = r/d_{mmed}$, where $r \in \{0.25, 1, 4\}$ and d_{mmed} is the median positive squared distance in a seeded subsample of 512 training rows; the implementation sets $d_{mmed} = 1$ if all those distances vanish. The kernel ridge grid is $\{10^{-4}, 10^{-2}, 10^{-1}\}$. All three regression heads have an unpenalized intercept and no coefficient-norm constraint. Together with two history-specific instances each of persistence and the constant predictor, these grids give 486 candidates per fit. All predictions are clipped to the fixed target interval $[0, 1]$.

Every model and selection rule is fixed before held-out scoring. The first execution evaluated calibration and test losses consecutively for each model, whereas the prescribed order completes all calibration calculations before test scoring. Repeating the calculations in the prescribed order gives identical selected predictors, group losses, and confidence endpoints; no choice depends on the intervening test losses. The two executions therefore represent one study. The source versions, evaluation-order records, and independent reconstruction are retained in the repository.

C.7 Reproduction and study provenance

The repository provides configurations, source hashes, candidate definitions, per-episode scores, and scripts for independently reconstructing the reported confidence endpoints. Episode-score archives support reconstruction; fixed-seed runners regenerate large files omitted from the repository. Archived environments specify numerical-library versions; bitwise agreement across architectures is not assumed. Complete commands for all studies, asset generation, and manuscript compilation are given in `extensions/certified_forecasting/docs/REPRODUCIBILITY.md`. Run-

ners preserve failed attempts and refuse to overwrite completed runs. Original-seed and confirmation input-scaling analyses remain separate, with the scope in Section 12.

C.8 Matched-geometry protocol

Each preprocessing arm has 72 matched scalar candidates and 12 controls. The refinement allocation covers 54 adjacent-prefix horizon comparisons: three geometries, two histories, three adjacent prefix pairs, and three horizons. The complete schedule and configuration hashes precede fitting; Section 13 specifies selection and joint confidence allocation. The result archive contains all 13,440 candidates, paired seed effects, horizon-specific results, and PenDigits writer comparisons. Table 7 combines absolute errors, fitting variation, and relative effects. The asset-generation scripts read these complete analysis records without refitting models or selecting configurations. Each study has a separate configuration and provenance record.

Appendix D. Exact-risk benchmark and centered comparator

This benchmark has an analytically known optimal risk and fitted-model risk, so certificate coverage can be checked without using a test estimate as population truth. Let $Z \in \{0, 1\}$ be a Bernoulli state with $\Pr(Z = 1) = p \in (0, 1)$, and let the scalar target be $Y = 2Z-1$. For a representation-separation parameter $a \in [0, 1]$, use

$$\Phi_a(0) = ((1 + a)/2, (1 - a)/2)^\top, \qquad \Phi_a(1) = ((1 - a)/2, (1 + a)/2)^\top.$$

Both vectors lie in the two-coordinate probability simplex. Their Euclidean distance is $\sqrt{2}a$, whereas the target separation is 2. For sensitivity budget $\kappa \geq 0$, define the unexplained separation $c = (2 - \kappa\sqrt{2}a)_+$. Proposition 3 then gives

$$R^* = p(1 - p)c^2.$$

Here R^* denotes the minimum MSE over the benchmark's constrained predictors. The two target values, representation family, and support are specified in advance by the benchmark design.

An independent training sample gives the empirical frequency $\widehat{p}$ of state 1. We choose the constrained empirical-risk minimizer whose predictions at the two states are

$$\widehat{f}(0) = -1 + \widehat{p}c, \qquad \widehat{f}(1) = 1 - (1 - \widehat{p})c. \tag{40}$$

These values lie in $[-1, 1]$ and have separation $2 - c = \min\{2, \kappa\sqrt{2}a\}$. The affine construction in Proposition 3 therefore realizes them within the budget; when $a = 0$, the two values coincide. If one state is absent from training, (40) selects one feasible empirical minimizer using the predeclared support. This convention specifies the prediction even at an unobserved state.

Conditional on the training sample, state 0 has prediction error $-\widehat{p}c$, and state 1 has error $(1-\widehat{p})c$. Consequently the exact fitted population risk is

$$R(\widehat{f}) = c^2\{(1 - p)\widehat{p}^2 + p(1 - \widehat{p})^2\} = R^* + c^2(\widehat{p} - p)^2. \tag{41}$$

For an independent pair of calibration states, the witness score equals $c^2/2$ when the states differ and zero otherwise. The mismatch probability is $2p(1 - p)$, so the expected score is exactly R^*. With n calibration observations, the pair lower bound uses $\lfloor n/2 \rfloor$ disjoint pairs. An odd final observation is excluded from pairing but remains in the fitted-risk and centered statistics. The reported calibration sizes are all even.

The centered comparator. For comparison, apply a centered certificate based on the target and representation variances, with Euclidean distance and a single reference class. Let $Z_1, \dots, Z_n$ be independent calibration states, with $n \geq 4$, and $\bar{z} = n^{-1} \sum_{i=1}^{n} Z_i$ their empirical frequency. The output and representation variances, computed with denominator n, are

$$\widehat{s} = 4\bar{z}(1-\bar{z}), \qquad \widehat{t} = 2a^2\bar{z}(1-\bar{z}).$$

Their population counterparts are $\sigma^2 = 4p(1-p)$ and $\tau^2 = 2a^2p(1-p)$. The population centered bound is $(\sigma - \kappa\tau)_+^2 \leq R^*$, using nonnegative square roots. In this two-state benchmark the inequality is an equality.

For a failure probability $\eta \in (0,1)$, define

$$\begin{aligned} r &= \log(2/\eta)/n, \\ V_\eta &= (\widehat{s} - 4\sqrt{r/2})_+, \\ T_\eta &= \min\{1, \widehat{t} + \sqrt{2r} + 1/n\}, \\ L_\eta^{\mathrm{ctr}} &= (\sqrt{V_\eta} - \kappa\sqrt{T_\eta})_+^2. \end{aligned} \tag{42}$$

Here V_η is a lower bound on σ^2, T_η is an upper bound on τ^2, and the superscript "ctr" identifies the centered risk bound. We show that $L_\eta^{\mathrm{ctr}} \leq R^*$ with probability at least $1-\eta$.

First, $\mathbb{E}\widehat{s} = (1-1/n)\sigma^2 \leq \sigma^2$, and changing one calibration observation changes $\widehat{s}$ by at most $4/n$. The bounded-differences inequality therefore gives

$$\widehat{s} - \mathbb{E}\widehat{s} \leq 4\sqrt{r/2}$$

with probability at least $1 - \eta/2$. On this confidence event, $V_\eta \leq \sigma^2$. Second, changing one observation changes $\widehat{t}$ by at most $2/n$, so the same inequality gives

$$\mathbb{E}\widehat{t} - \widehat{t} \leq \sqrt{2r}$$

with failure probability at most $\eta/2$. Since $\mathbb{E}\widehat{t} = (1-1/n)\tau^2$ and $\tau^2 \leq 1$,

$$\tau^2 \leq \widehat{t} + \sqrt{2r} + \tau^2/n \leq \widehat{t} + \sqrt{2r} + 1/n.$$

Capping the right side at one remains valid because $\tau^2 \leq 1$. A union bound gives $V_\eta \leq \sigma^2$ and $T_\eta \geq \tau^2$ jointly with probability at least $1-\eta$. The expression in (42) increases with its first variance argument and decreases with its second. Hence it is no larger than $(\sigma - \kappa\tau)_+^2 \leq R^*$ on that event. The constant one is the general simplex-variance bound; it is conservative for this two-coordinate example.

In each Monte Carlo trial, the fitted-risk upper bound, pair-witness lower bound, and centered lower bound each receive failure probability $\delta/3$. For the centered bound, setting $\eta = \delta/3$ splits that allocation between its two internal variance bounds. The three reported directions therefore hold jointly with probability at least $1-\delta$ within a trial. No joint $1-\delta$ event is asserted across all trials, grid conditions, or separately reported experiments.

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